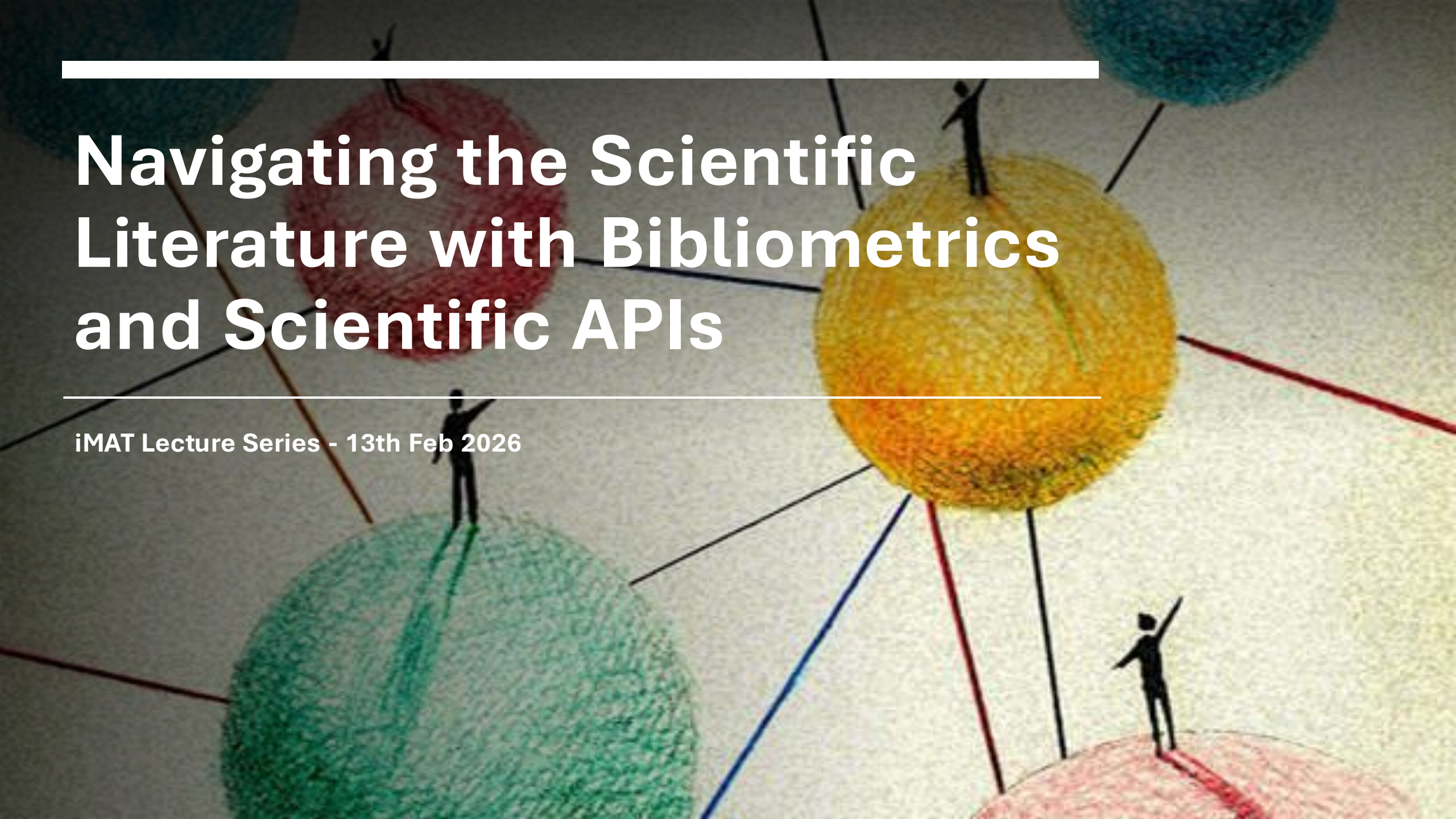


An abstract background featuring several large, textured spheres in red, yellow, green, and blue. Thin black lines connect these spheres, and small black human figures are standing on some of the spheres, suggesting a network or a journey. A thick white horizontal bar is at the top left.

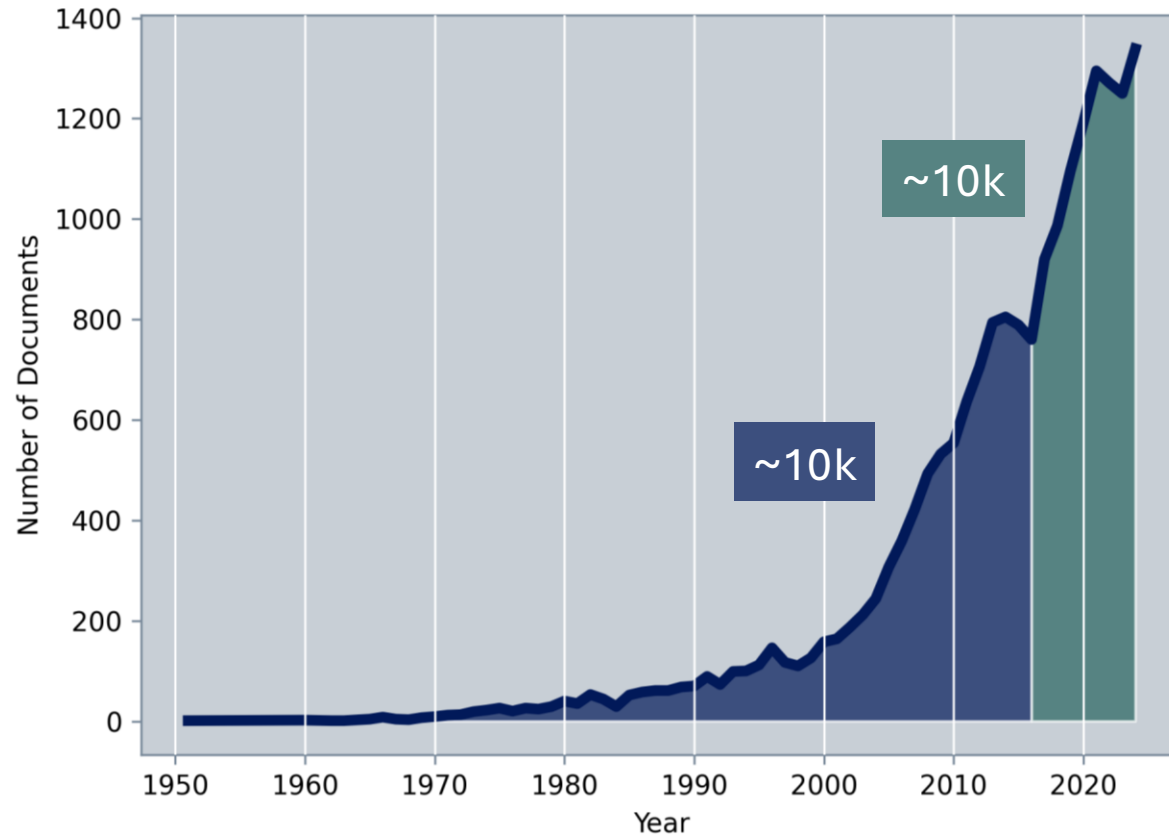
Navigating the Scientific Literature with Bibliometrics and Scientific APIs

iMAT Lecture Series - 13th Feb 2026

“Much of a scientist’s work consists of reading and writing.”

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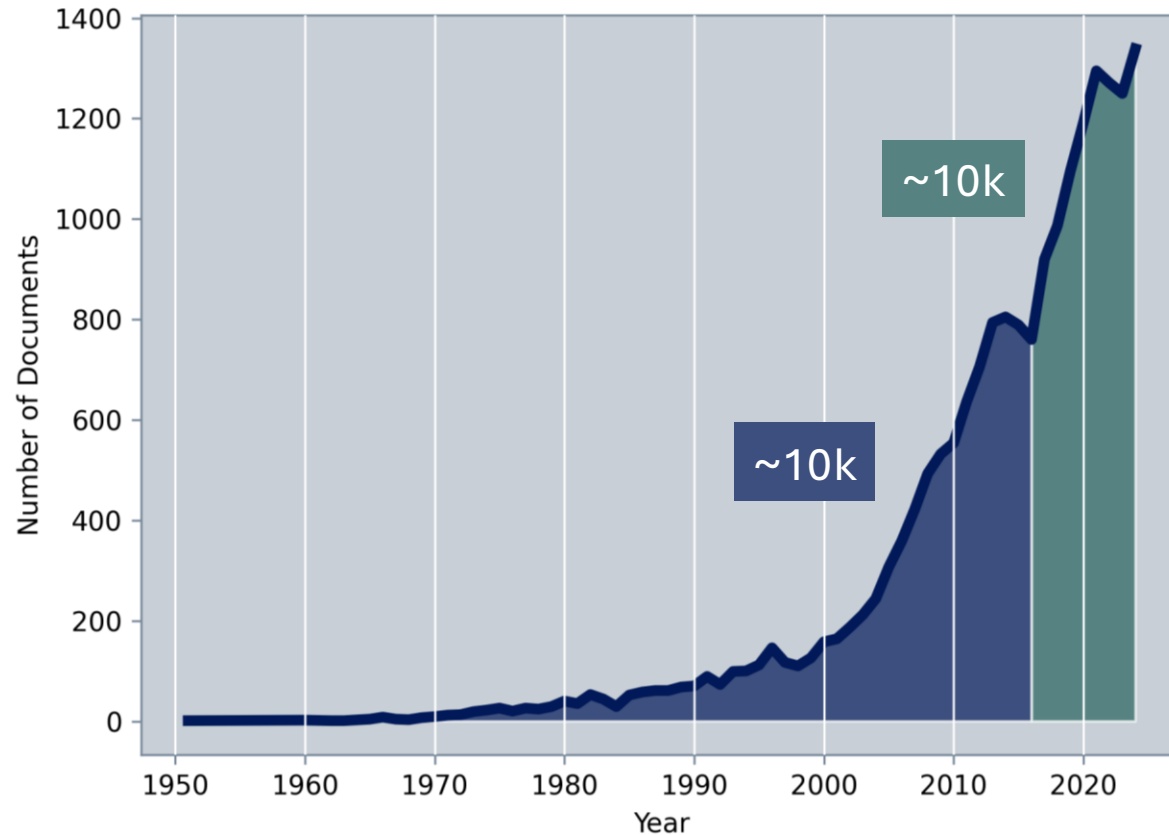
“Much of a scientist’s work consists of **reading** and writing.”



The number of publications published in the last 10 years equals the number of publications from 1951 to 2016¹

¹Pradelli et al. (2026). 75 Years of Mathematical Oncology <https://doi.org/10.64898/2026.01.13.6993.06>

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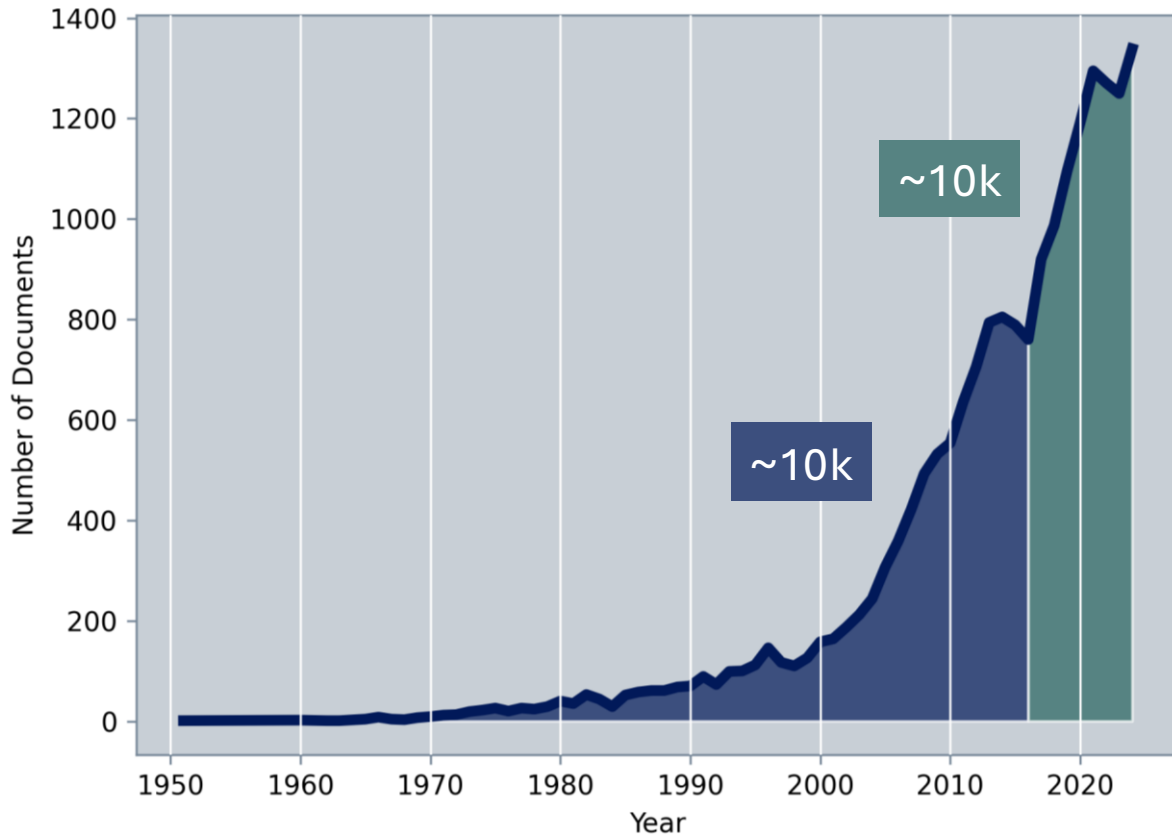
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Increasingly difficult to:

- **Stay updated**
- **Learn new things**

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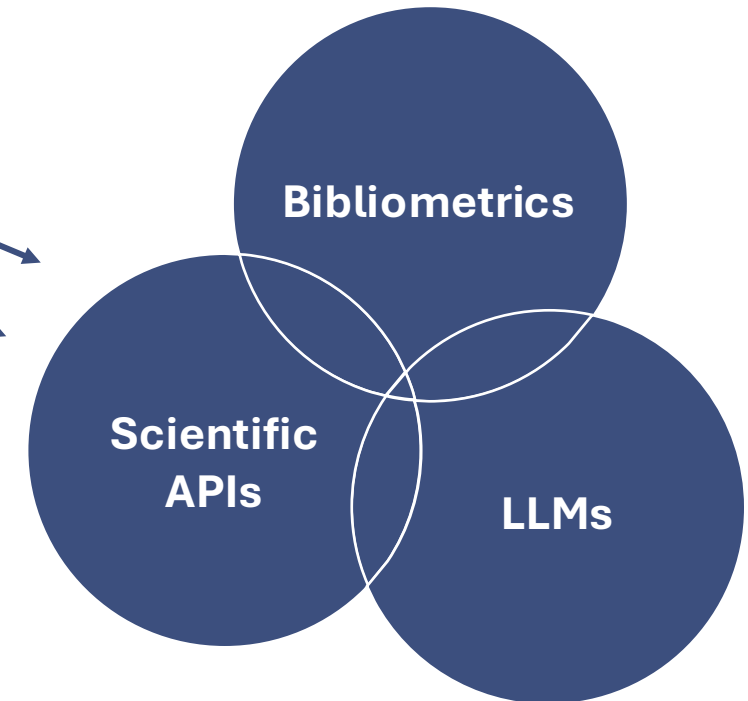
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- Introduction
- Staying updated with web scraping
- Exploring the literature with Bibliometrics
- Integrations with Large Language Models

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Some definitions



Bibliometric Data

Bibliometric Analysis

Scientific APIs

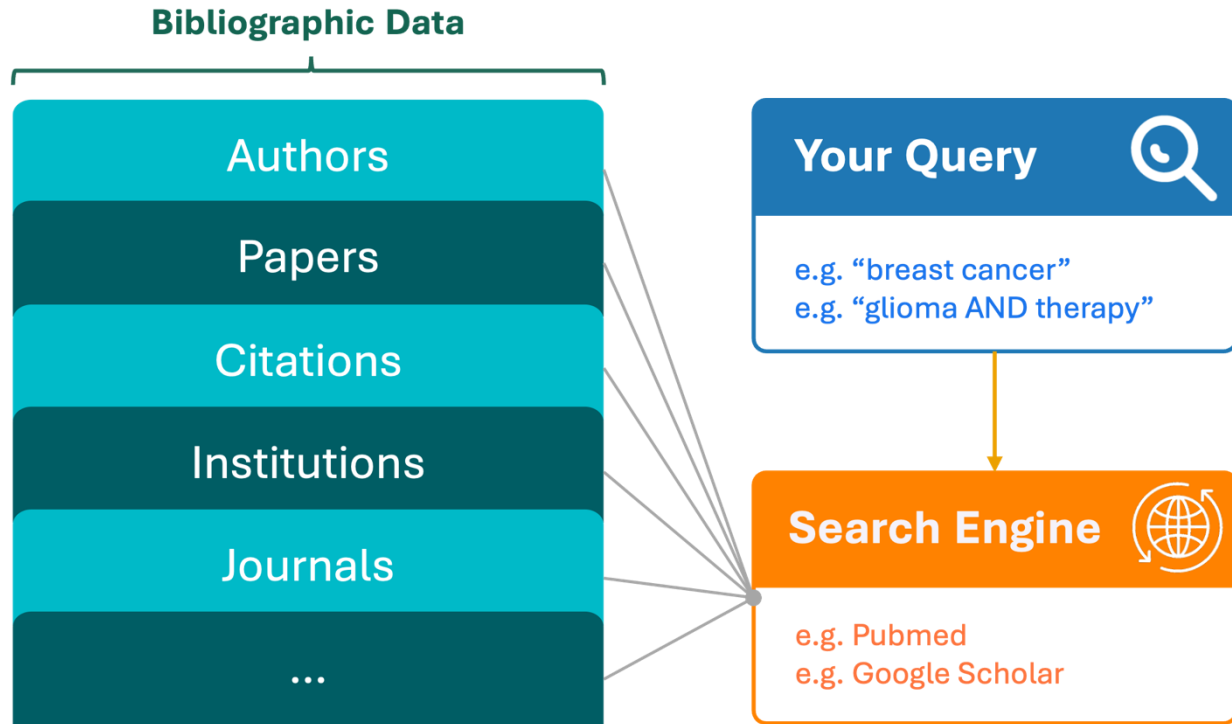
Some definitions



Bibliometric Data

Bibliometric Analysis

Scientific APIs



Some definitions



Bibliometric Data

Bibliometric Analysis

Scientific APIs

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

PubMed®

"Mathematical Model" AND "Cancer"

Advanced Create alert Create RSS User Guide

Save Email Send to Sort by: Best match Display options

MY CUSTOM FILTERS

3,123 results

Page 1 of 313

RESULTS BY YEAR

1966 2026

PUBLICATION DATE

1 year
5 years
10 years
Custom Range

TEXT AVAILABILITY

Abstract
Free full text
Full text

ARTICLE ATTRIBUTE

Associated data

1
Cite

2
Cite

Mathematical Model of Cancer Treatment with Virotherapy and Immune System.
Sinha AK, Sahu A.
Crit Rev Biomed Eng. 2025;53(3):1-11. doi: 10.1615/CritRevBiomedEng.2025055125.
PMID: 40749192 Review.

The global burden of **cancer** is rising, causing significant strain on individuals, families, and healthcare systems. ...In contrast, virotherapy specifically targets **cancer** cells, leaving healthy cells unharmed. This study presents a **mathematical model** ...

Papers PDF

Evaluating cancer etiology and risk with a mathematical model of tumor evolution.
Pénisson S, Lambert A, Tomasetti C.
Nat Commun. 2022 Nov 24;13(1):7224. doi: 10.1038/s41467-022-34760-1.
PMID: 36433937 Free PMC article.

Download PDF

A deeper understanding of the evolution of this endogenous mutational load is critical for understanding what causes **cancer**. Here we present a **mathematical model** of tumor evolution that is able to predict the expected number of endogenous somatic mutations pr ...

In Library

Lung Cancer Radiotherapy: Simulation and Analysis Based on a

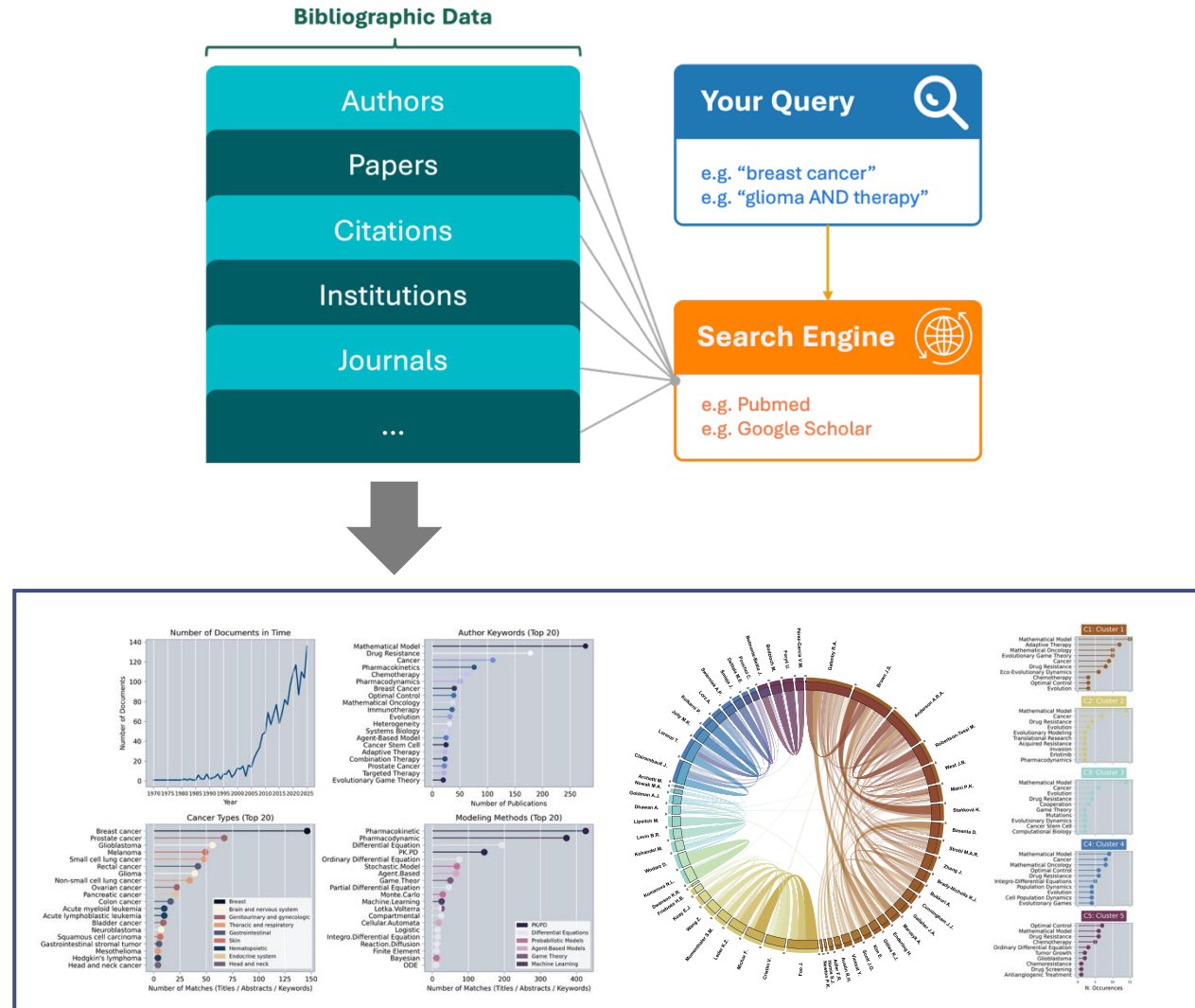
Some definitions



Bibliometric Data

Bibliometric Analysis

Scientific APIs



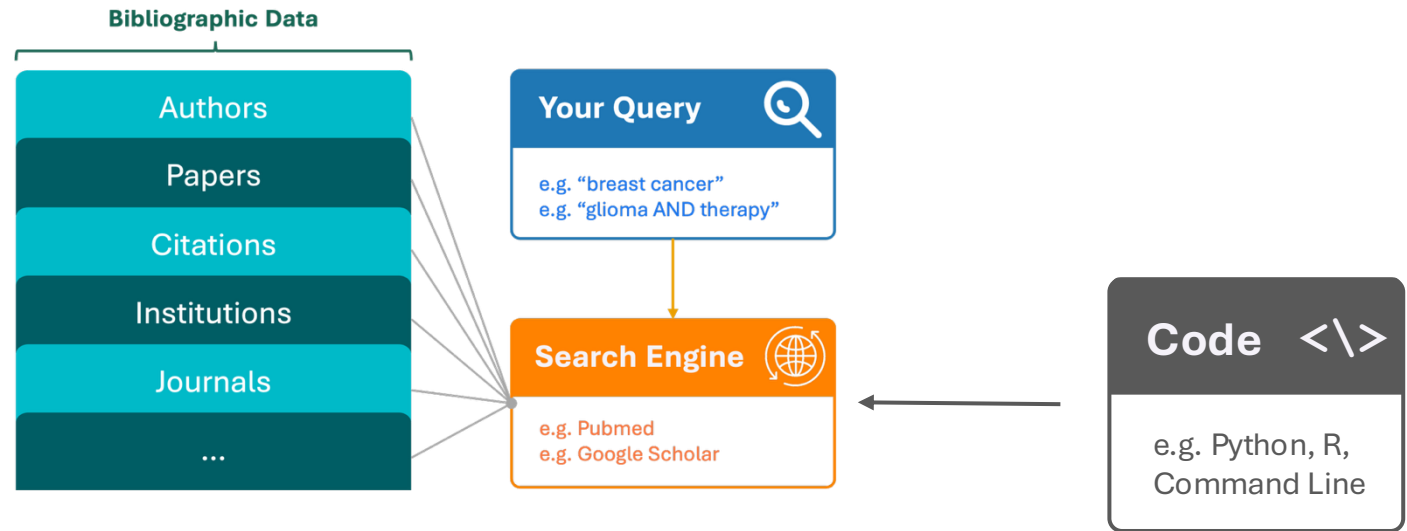
Some definitions



Bibliometric Data

Bibliometric Analysis

Scientific APIs



Some definitions



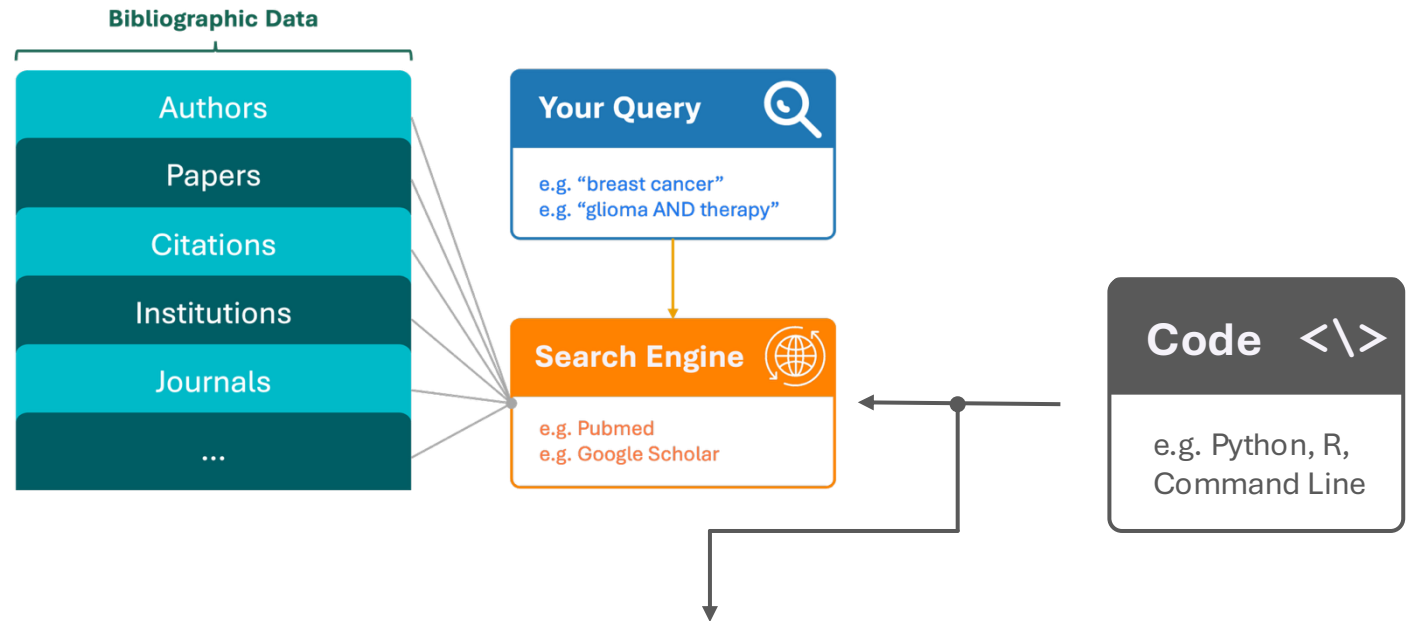
Bibliometric Data

Bibliometric Analysis

Scientific APIs

Some examples:

- Scopus: <https://dev.elsevier.com/documentation/ScopusSearchAPI.wadl>
- PubMed: <https://www.ncbi.nlm.nih.gov/home/develop/api/>
- OpenAlex: <https://docs.openalex.org/how-to-use-the-api/api-overview>
- Google Scholar (SerpAPI): <https://serpapi.com>



API = Application Programming Interface

RESTful API = Standard for **web interfaces** (http domains)

```
> curl https://official-joke-api.appspot.com/random_joke
{"type":"general","setup":"Why does Norway have barcodes on their battleships?",
"punchline":"So when they get back to port, they can Scandinavian.","id":359}
```

```
import requests

response = requests.get("https://api.example.com/users/42")
data = response.json()

print(data["name"])
```

Some definitions



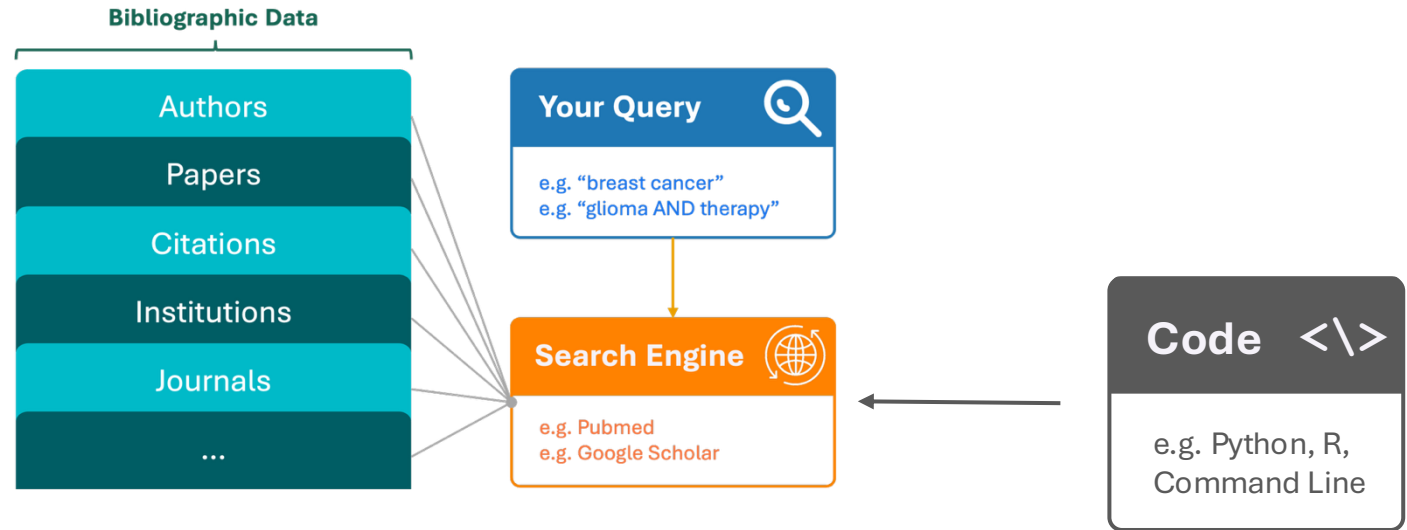
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- PubMed: <https://www.ncbi.nlm.nih.gov/home/develop/api/>
- OpenAlex: <https://docs.openalex.org/how-to-use-the-api/api-overview>
- Google Scholar (SerpAPI): <https://serpapi.com>



Scopus Search API

This represents the search interfaces associated with Scopus abstracts. Each search result will, by definition, link to a [Scopus abstract](#). Search entries might also have links to a [Full-Text article](#) representation of the result.

Additional information includes [Scopus Search Views](#) and [Scopus Search Tips](#).

Summary

Resource	Method	Description
https://api.elsevier.com/content/search/scopus	GET	SCOPUS Search API: This represents a search against the SCOPUS cluster, which contains SCOPUS abstracts. Scopus is the largest abstract and citation database of research literature and quality web sources. Updated daily, Scopus covers 50 million abstracts of over 20,500 peer-reviewed titles from more than 5,000 publishers. This search resource allows for the submission of Boolean queries into the Scopus index, retrieving relevant result metadata in a user-specific text formats.

Resources

<https://api.elsevier.com/content/search/scopus>

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"LiRA: Literature Review Automated"

🔗 Automate your weekly scientific literature review

LiRA is a CLI Python program based on PyMed and SerpAPI to search on PubMed and get the results programmatically in a readable HTML page.

I created LiRA mainly for myself, but feel free to use it if you find it useful.



<https://github.com/fpradelli94/LiRA>

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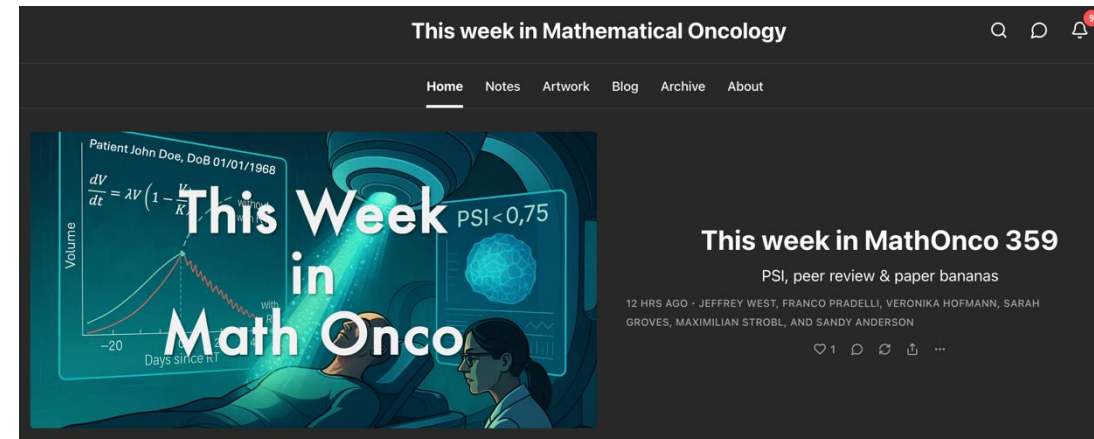
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Subscribe: <https://thisweekmathonco.substack.com/about>

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1. Clone the repository
2. Install dependencies
3. **Configuration**

```
{
  "engines": ["pubmed", "google_scholar"],
  "email": "example@gmail.com",
  "serpapi_key": "aobosandi392309qwjadossasioiq"
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    "Breast Cancer",
    "Omics",
    "cancer AND immunology"
  ],
  "authors": [
    "Kaelin, William",
    "Doudna, Jennifer"
  ],
  "journals": [
    "ArXiv",
    "bioRxiv",
    "Nature"
  ],
  "highlight_authors": [
    "Doe, John"
  ]
}
```

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```
pipenv run python3 src/lira.py --from-date 2023/09/19  
# OR  
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```

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General results [pubmed](34) (2026/02/04 - 2026/02/11)

Show paragraphs of text ☒

Community assembly modeling of the microbiome within Barrett's esophagus and esophageal adenocarcinoma.

Guccione, Caitlin et al. (2026/02/11)
<https://pubmed.ncbi.nlm.nih.gov/41667953>

None

TNF- α -mediated myeloid-instructed CD14⁺CD4⁺ T cells are associated with poor survival in lung adenocarcinoma.

Marceaux, Claire et al. (2026/02/11)
<https://pubmed.ncbi.nlm.nih.gov/41666923>

The tumor microenvironment is composed of diverse immune populations that can either support anti-tumor immunity or promote tumor progression. Myeloid cells are major drivers of immunosuppression, yet therapies targeting them have shown limited success. To uncover mechanisms underlying myeloid-driven immune suppression, we performed spatial multi-omics analyses of non-small cell lung cancer (NSCLC). Independent of oncogenic driver status, tumors stratify into lymphoid-enriched, myeloid-enriched, and mixed immune-infiltrated subtypes. In tumor and adjacent non-malignant lungs, we identify myeloid-instructed CD14⁺CD4⁺ T cells. These cells arise through trogocytosis adopting an atypical phenotype. In lymphoid-enriched tumors, high infiltration of CD14⁺CD4⁺ T cells correlates with poor patient survival. Spatial transcriptomics reveal enrichment of tumor necrosis factor alpha (TNF- α) signaling in CD14⁺CD4⁺-T-cell-rich tumors. Functional assays demonstrate that TNF- α enhanced trogocytosis, promoting the formation of CD14⁺CD4⁺ T cells. These findings uncover a TNF- α -mediated mechanism of immunosuppression in the TME and highlight aberrant myeloid-T cell interactions as contributors to NSCLC progression.

Matched Analysis of the Risk Assessment and Prediction Tool for Post-Operative Disposition Needs in a Spinal Oncology Population.

Xu, Emily et al. (2026/02/10)
<https://pubmed.ncbi.nlm.nih.gov/41666294>

Study DesignRetrospective cohort study.ObjectivesAs cancer survival improves, metastatic spinal cancer has become increasingly common worldwide. Given the high resource demands of

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The '-w' options allows you to search in the past weeks

```
pipenv run python3 src/lira.py --for-weeks 2  
# OR  
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```

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New Results

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75 Years of Mathematical Oncology

Franco Pradelli, Maximilian Strobl, Sadegh Marzban, Francois de Kermenguy, Ari Barnett, Katayni Ganesan, Guillermo Lorenzo, David A. Hormuth II, Sara Hamis, Dhananjay Bhaskar, Alexander R. A. Anderson, Jeffrey West

doi: <https://doi.org/10.64898/2026.01.13.699306>

This article is a preprint and has not been certified by peer review [what does this mean?].

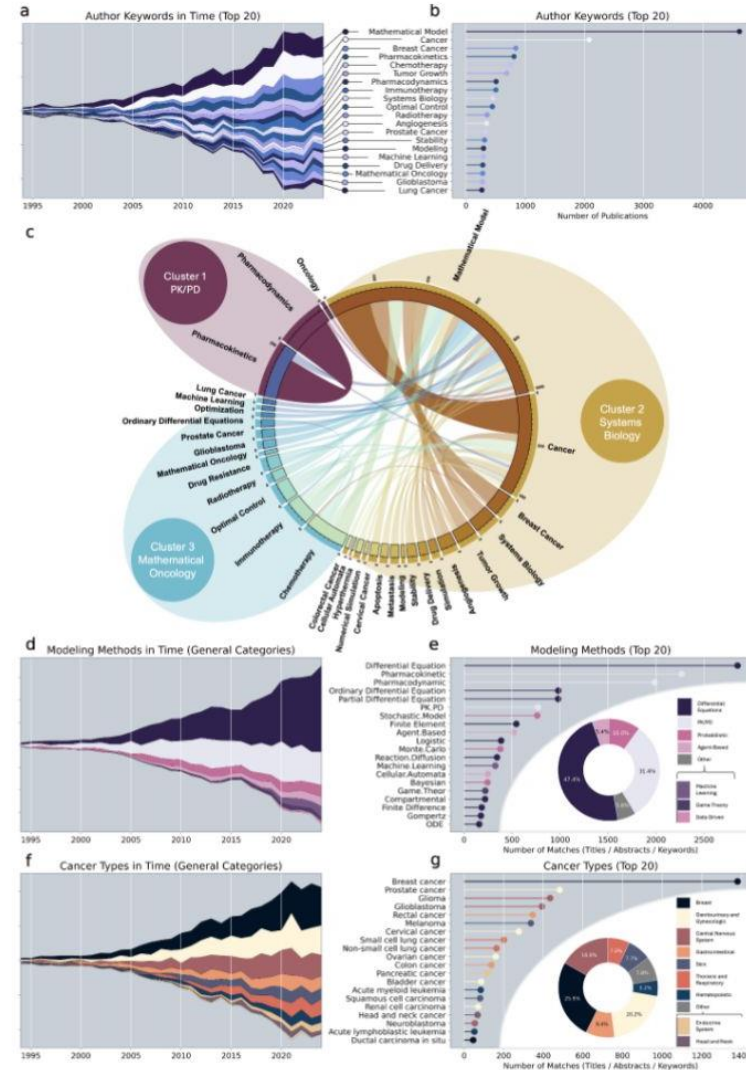
Abstract

[Info/History](#) [Metrics](#)

[Preview PDF](#)

Abstract

Mathematics has long provided a quantitative framework to interpret cancer biology and treatment. Driven by richer biological and clinical data, the field has evolved into a multidisciplinary and translational endeavor - giving rise to Mathematical Oncology. Yet, the field's strong interdisciplinarity obscures a comprehensive view of its evolution, as well as the boundaries between Mathematical Oncology and adjacent domains. Here, we address this gap through a comprehensive bibliometric analysis spanning 75 years of mathematical research in oncology. Using a manually curated corpus (~1,500 papers) and a large query-based dataset (~19,000 papers), we map the field's conceptual and collaborative development over time. Independent analyses reveal sustained growth, high impact, and pronounced interdisciplinarity, together with a gradual shift from fundamental cancer biology toward therapeutic modeling. This reorientation underpins the emergence of Mathematical Oncology as a distinct field, separate from Systems Biology and Pharmacokinetics / Pharmacodynamics. We show how diverse concepts interlace within Mathematical Oncology, bridging applied mathematics, optimal control, evolutionary theory, and imaging. Collectively, we demonstrate that Mathematical Oncology is not merely the application of mathematics to cancer, but the use of interpretable models integrating clinical, biological, and physical knowledge to improve screening, understand disease evolution, guide therapy, and strengthen forecasting.



75 Years of Mathematical Oncology

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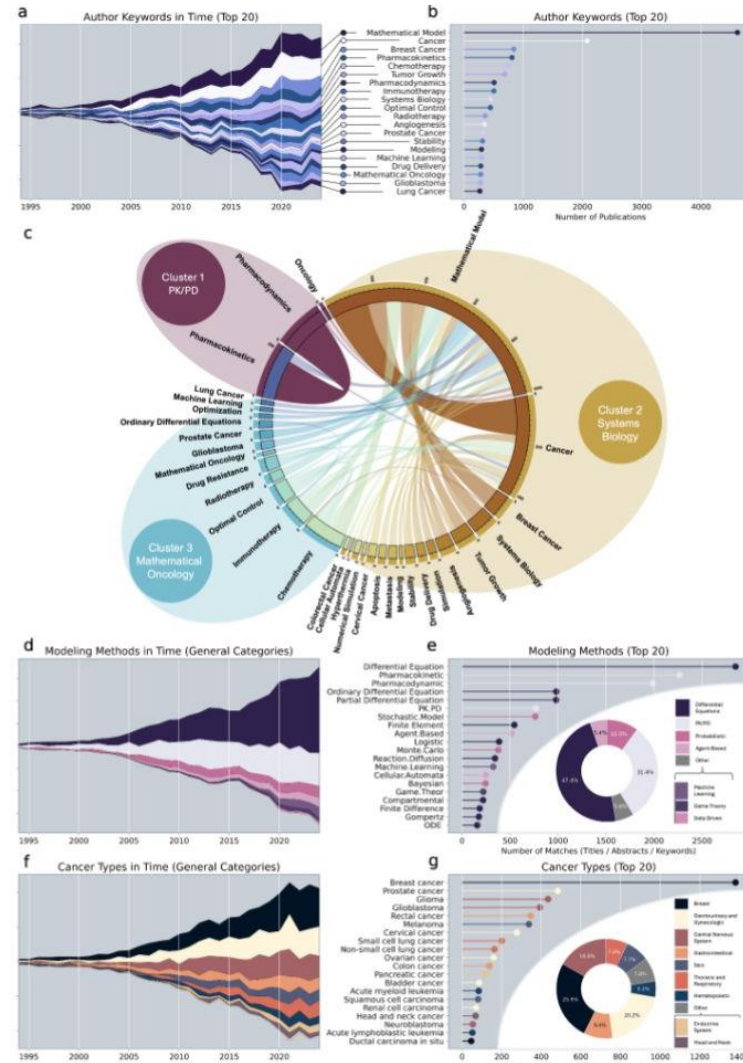
Abstract

Info/History Metrics

Preview PDF

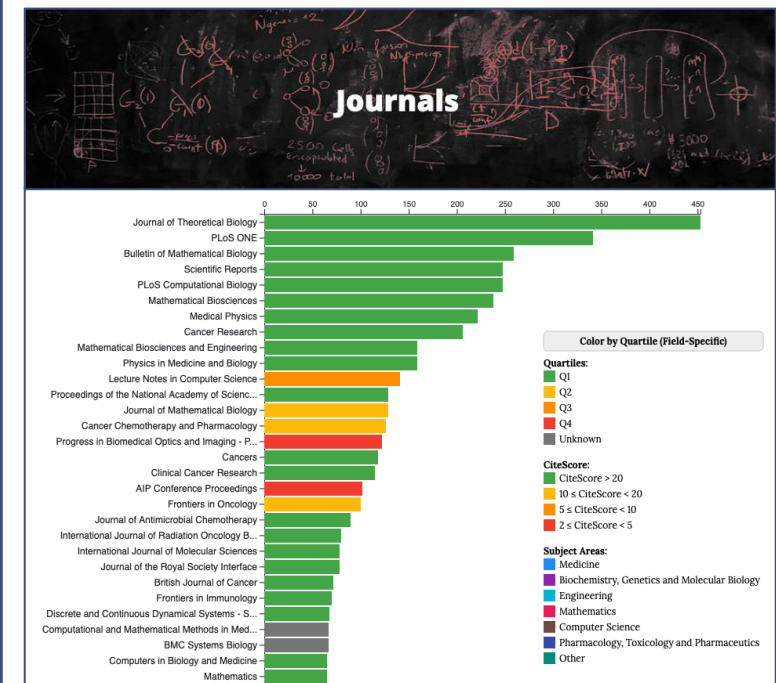
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Know better our field

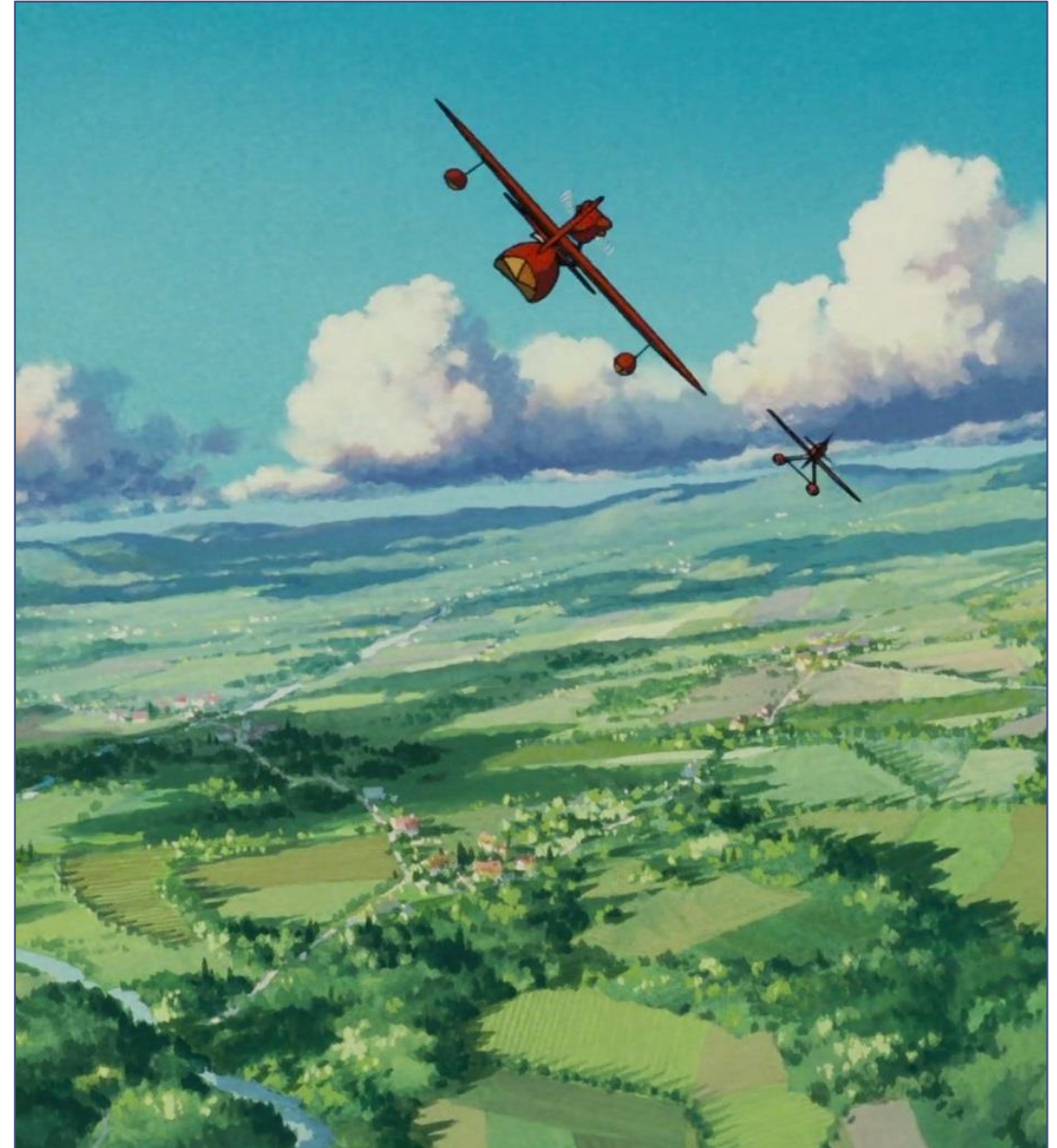
Create useful tools for
Mathematical Oncologists



Link: <https://mathematical-oncology.org/journals>

How can it be useful?

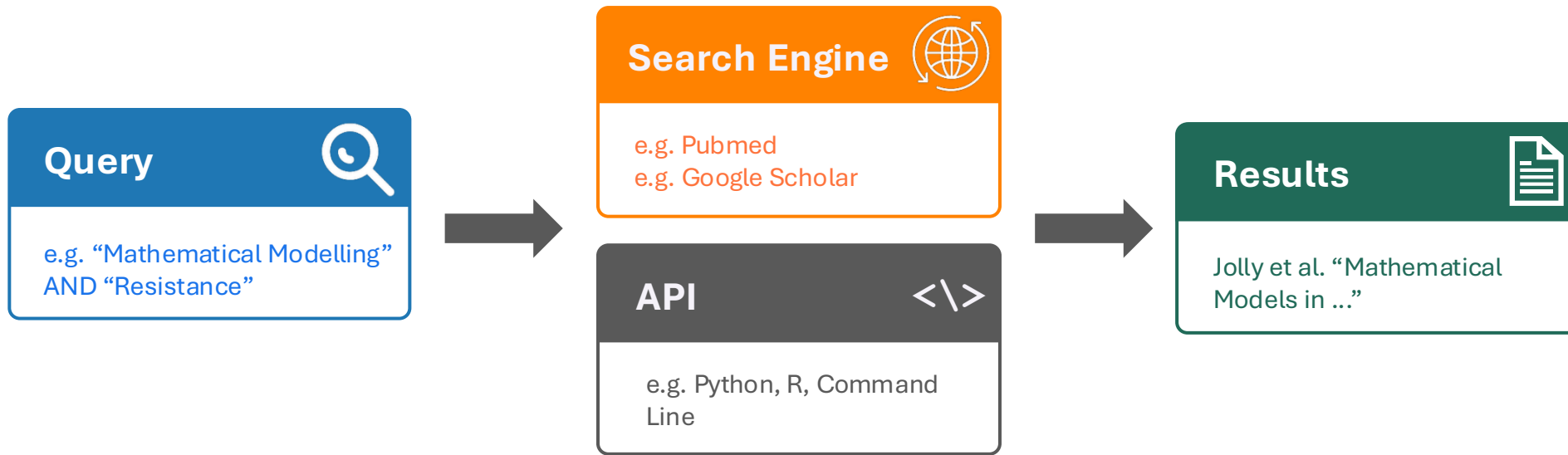
- **Coarse-grained** but **quantitative**
- Prioritizing what to read
- General concepts
- Research papers faster and systematically



Example: Mathematical Modelling in Cancer Resistance²

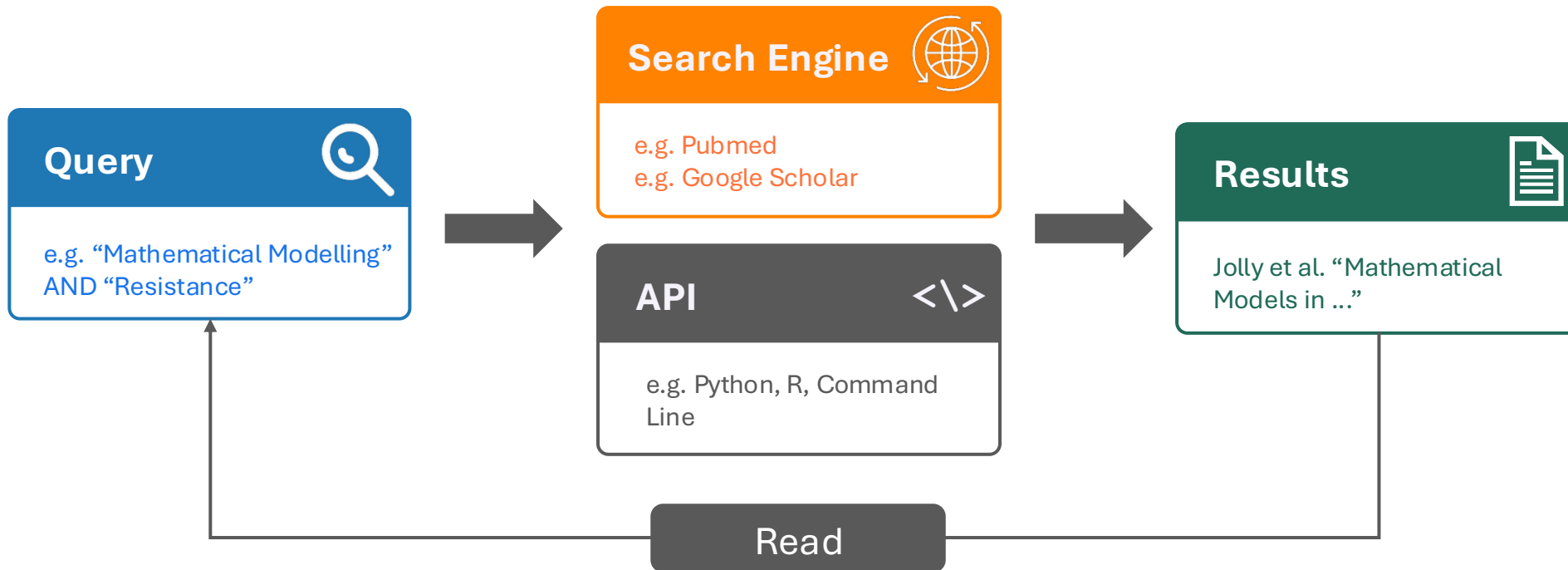
²Pradelli F. & West J (2026). Mathematical Models of Cancer Resistance. Book Chapter for Molecular Mechanisms of Cancer Resistance (Elsevier). Submitted.

Example: Mathematical Modelling in Cancer Resistance²



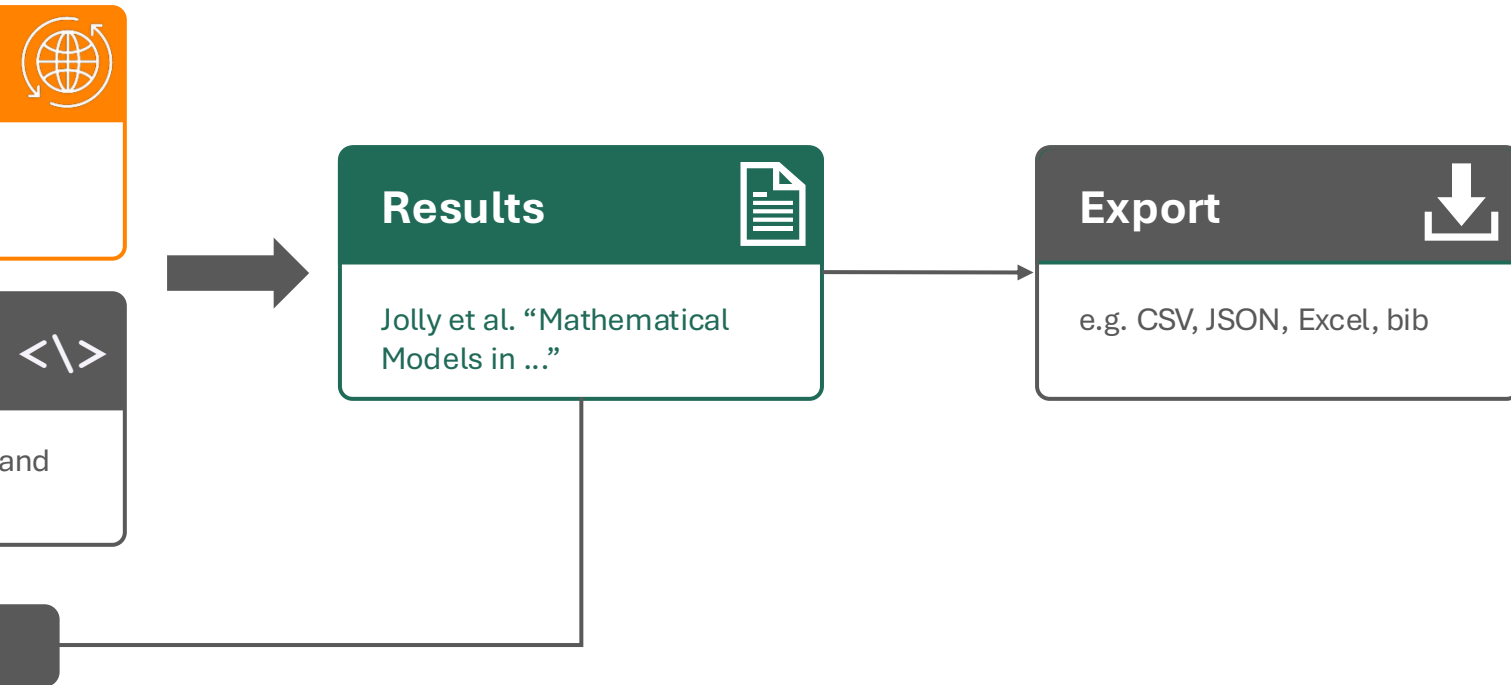
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Example: Mathematical Modelling in Cancer Resistance²



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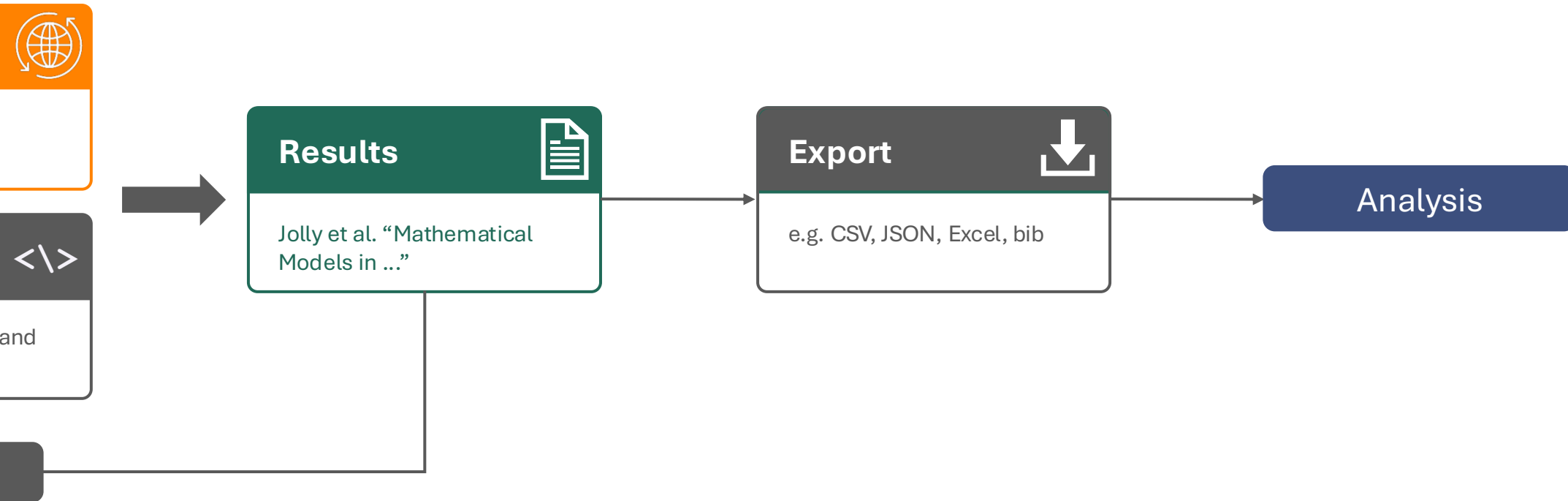
Example: Mathematical Modelling in Cancer Resistance²



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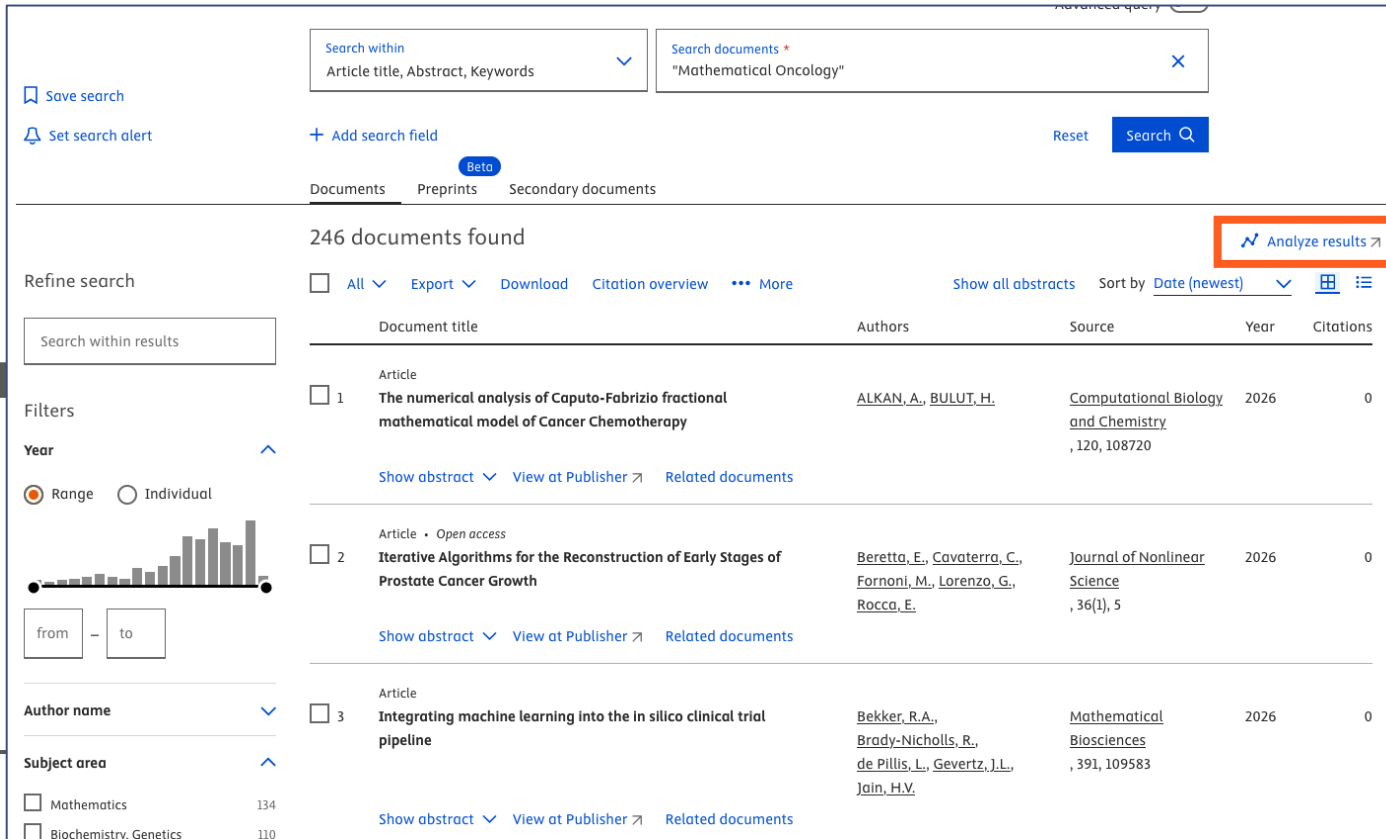
and

Example: Mathematical Modelling in Cancer Resistance²



²Pradelli F. & West J (2026). Mathematical Models of Cancer Resistance. Book Chapter for Molecular Mechanisms of Cancer Resistance (Elsevier). Submitted.

Example: Mathematical Modelling in Cancer Resistance²



The screenshot shows a search interface with the following elements:

- Search bar:** "Search documents" with the query "Mathematical Oncology".
- Buttons:** "Save search", "Set search alert", "Add search field", "Reset", "Search", and "Analyze results" (highlighted with a red box).
- Results:** 246 documents found. The results are displayed in a table with columns: Document title, Authors, Source, Year, and Citations.
- Filters:** A sidebar on the left includes a "Refine search" box, a "Search within results" box, and a "Filters" section with a "Year" histogram and "Author name" and "Subject area" dropdowns.
- Document List:**
 - 1. **The numerical analysis of Caputo-Fabrizio fractional mathematical model of Cancer Chemotherapy** by ALKAN, A., BULUT, H., published in *Computational Biology and Chemistry*, 2026, 120, 108720.
 - 2. **Iterative Algorithms for the Reconstruction of Early Stages of Prostate Cancer Growth** (Open access) by Beretta, E., Cavaterra, C., Fornoni, M., Lorenzo, G., Rocca, E., published in *Journal of Nonlinear Science*, 2026, 36(1), 5.
 - 3. **Integrating machine learning into the in silico clinical trial pipeline** by Bekker, R.A., Brady-Nicholls, R., de Pillis, L., Gevertz, J.L., Jain, H.V., published in *Mathematical Biosciences*, 2026, 391, 109583.

Analysis

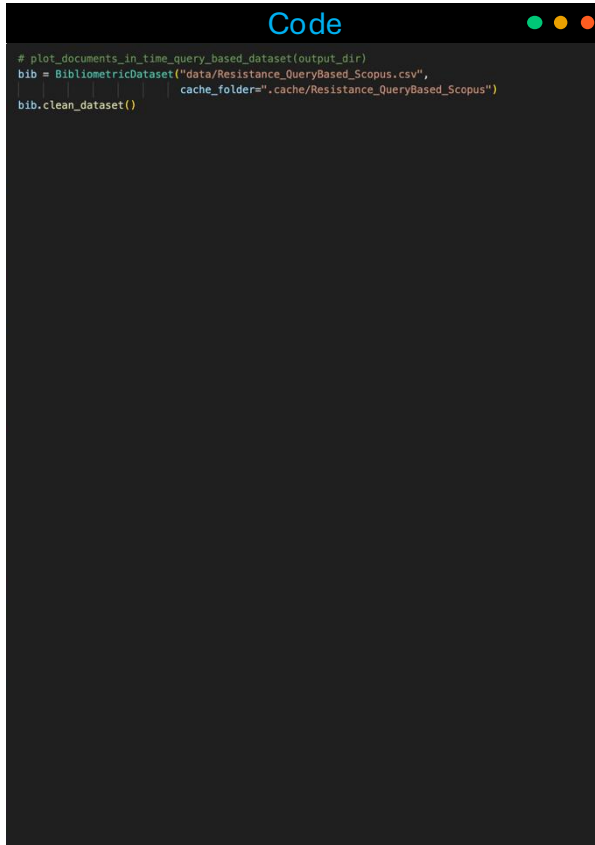
I use my own code (Python) + VosViewer

Other resources (R): <https://www.bibliometrix.org/>

²Pradelli F. & West J (2026). Mathematical Models of Cancer Resistance. Book Chapter for Molecular Mechanisms of Cancer Resistance (Elsevier). Submitted.

Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance



```
# plot_documents_in_time_query_based_dataset(output_dir)
bib = BibliometricDataset("data/Resistance_QueryBased_Scopus.csv",
                          cache_folder=".cache/Resistance_QueryBased_Scopus")
bib.clean_dataset()
```

Cleaning the dataset from³:

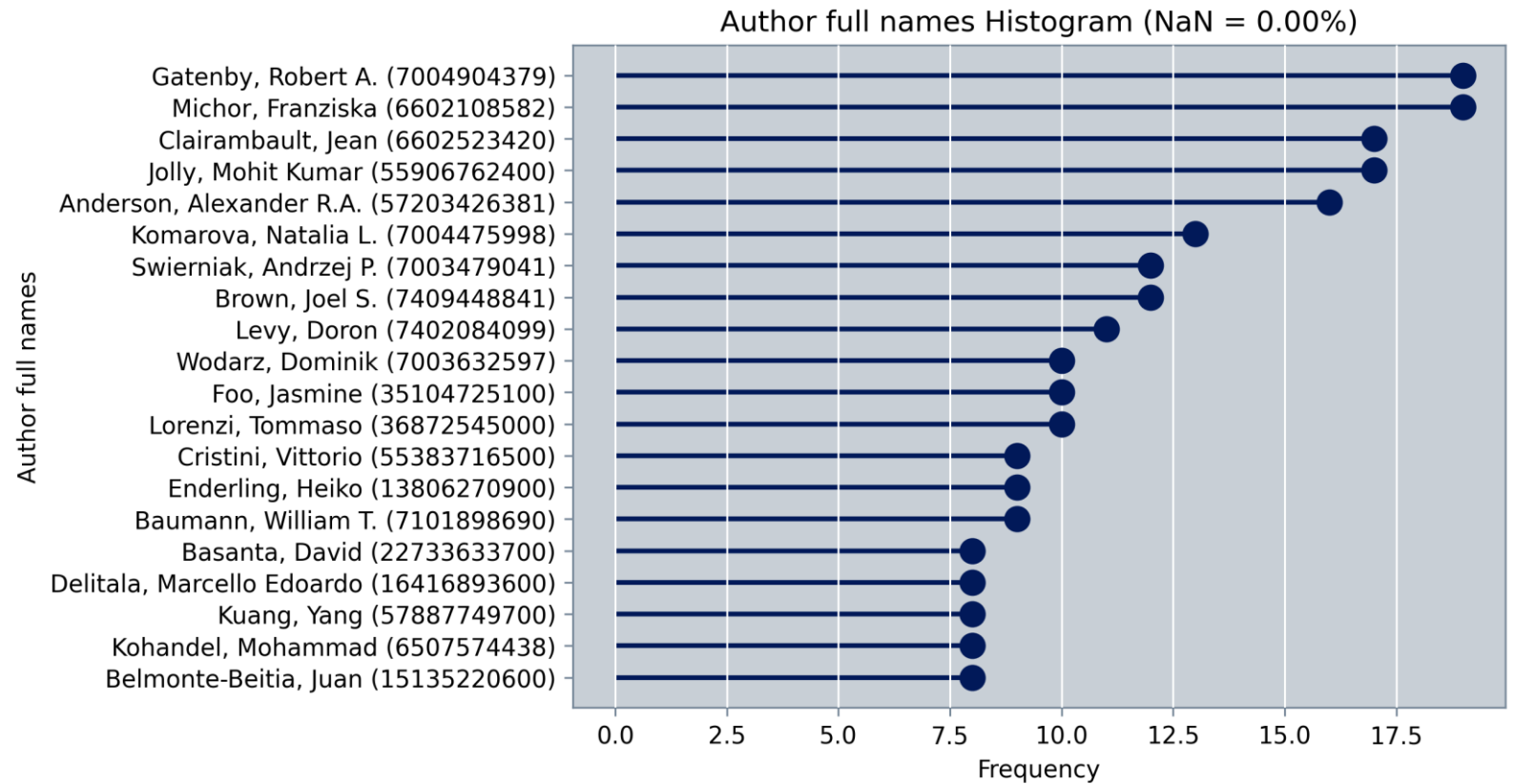
- Duplicate entries
- NaNs
- Spelling mistakes

Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code
# plot_documents_in_time_query_based_dataset(output_dir)
bib = BibliometricDataset("data/Resistance_QueryBased_Scopus.csv",
                           cache_folder=".cache/Resistance_QueryBased_Scopus")
bib.clean_dataset()

# plot authors
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_authors_ranking(top_n=20, ex=axes)
fig.savefig(output_dir / "authors_ranking.png", bbox_inches='tight',
            dpi=300)
fig.tight_layout()
```



Bibliometrics: see authors

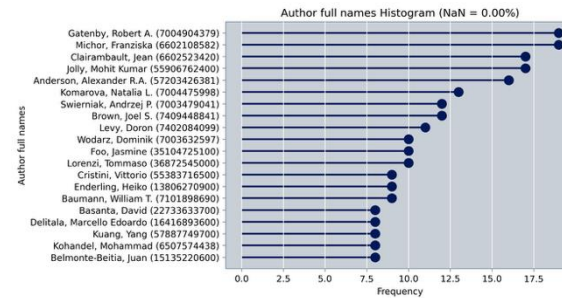
Example: Mathematical Modelling in Cancer Resistance

```
Code
# plot_documents_in_time_query_based_dataset(output_dir)
bib = BibliometricDataset("data/Resistance_QueryBased_Scopus.csv",
                           cache_folder=".cache/Resistance_QueryBased_Scopus")
bib.clean_dataset()

# plot authors
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "authors_ranking.png", bbox_inches='tight',
            dpi=300)
fig.tight_layout()

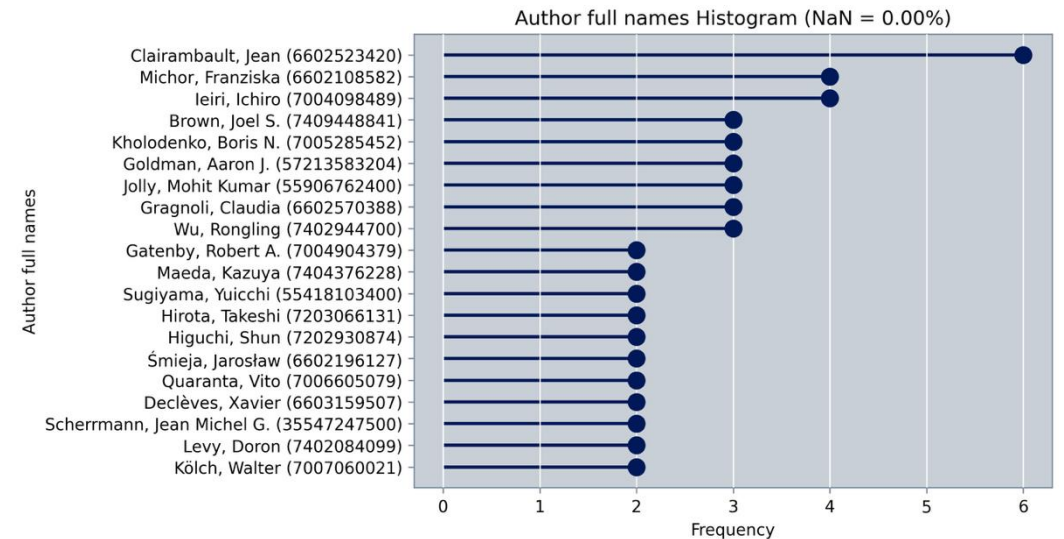
# do the same selecting reviews
bib_reviews = bib[bib["Document Type"] == "Review"]

fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib_reviews.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "authors_ranking_reviews.png", bbox_inches='tight',
            dpi=300)
```



Full dataset

Only Reviews



Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

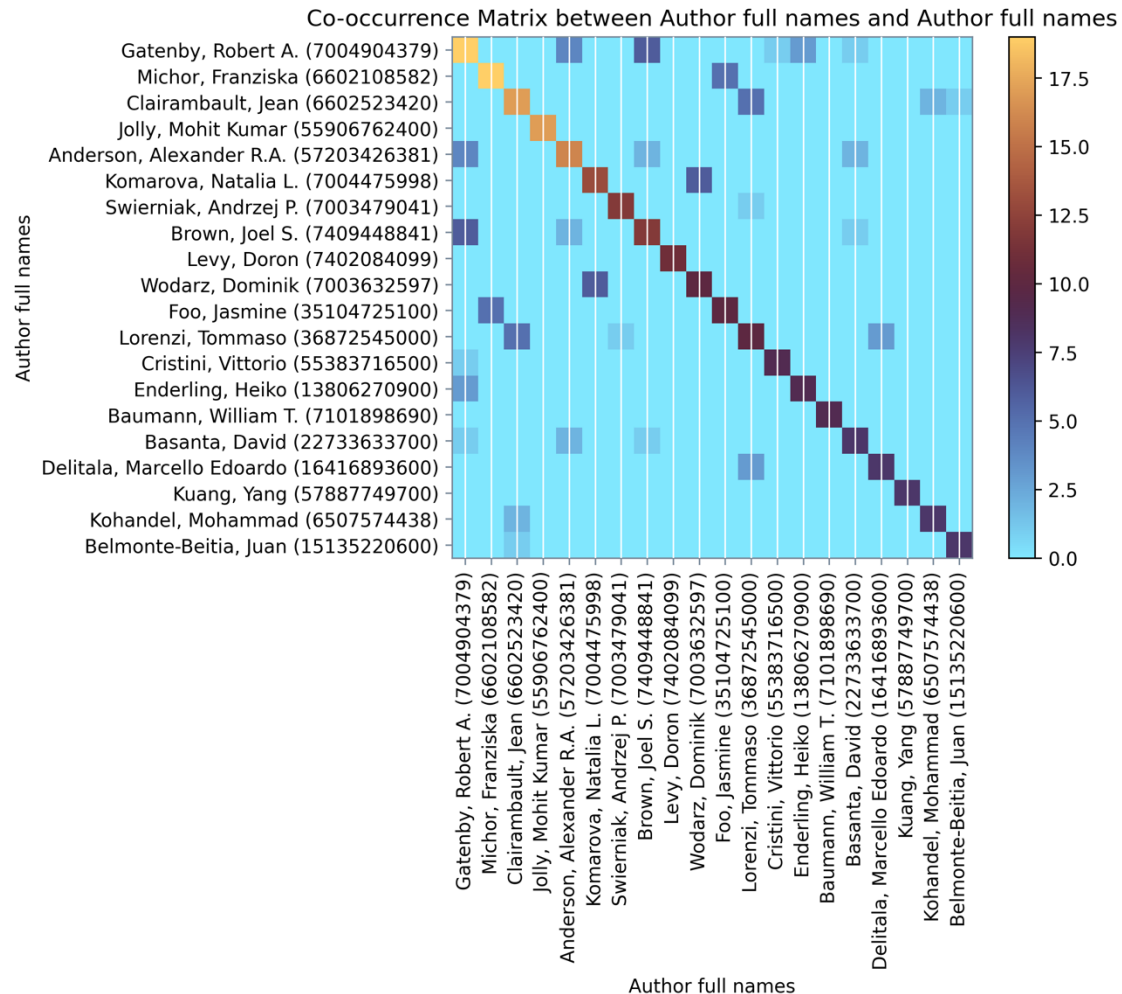
```
Code
# plot_documents_in_time_query_based_dataset(output_dir)
bib = BibliometricDataset("data/Resistance_QueryBased_Scopus.csv",
                           cache_folder=".cache/Resistance_QueryBased_Scopus")
bib.clean_dataset()

# plot authors
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
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fig.savefig(output_dir / "authors_ranking.png", bbox_inches='tight',
            dpi=300)
fig.tight_layout()

# do the same selecting reviews
bib_reviews = bib[bib["Document Type"] == "Review"]

fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib_reviews.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "authors_ranking_reviews.png", bbox_inches='tight',
            dpi=300)

# see correlation between authors
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                             column_2="Author full names",
                                             top_n1=20,
                                             top_n2=20,
                                             ax=axes)
fig.savefig(output_dir / "authors_cooccurrence.png", bbox_inches='tight',
            dpi=300)
```



Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code
# plot_documents_in_time_query_based_dataset(output_dir)
bib = BibliometricDataset("data/Resistance_QueryBased_Scopus.csv",
                           cache_folder=".cache/Resistance_QueryBased_Scopus")
bib.clean_dataset()

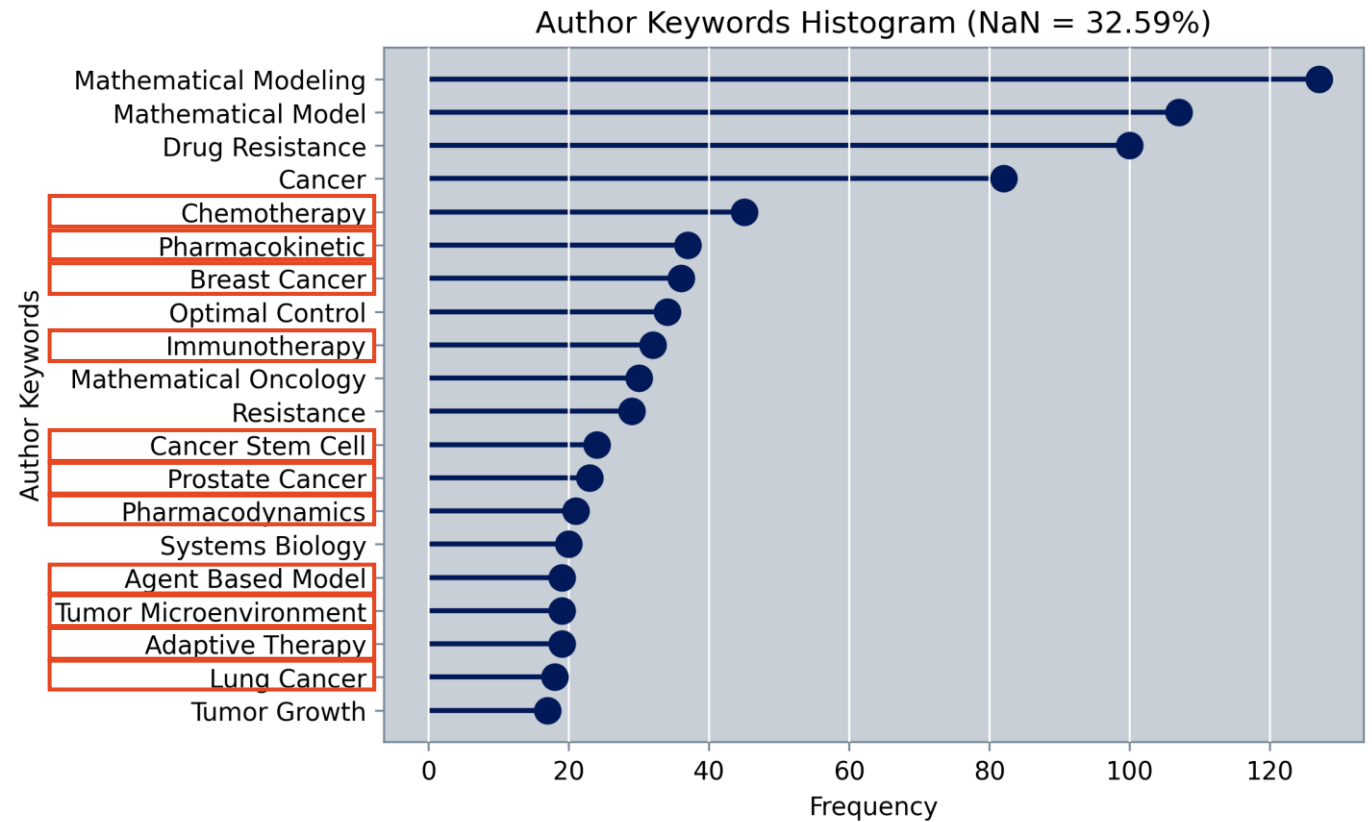
# plot authors
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "authors_ranking.png", bbox_inches='tight',
            dpi=300)
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bib_reviews.plot_authors_ranking(top_n=20, ax=axes)
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fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                             column_2="Author full names",
                                             top_n1=20,
                                             top_n2=20,
                                             ax=axes)
fig.savefig(output_dir / "authors_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# see author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.clean_author_keywords(threshold=0.9)
bib.plot_author_keywords_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "author_keywords_ranking.png", bbox_inches='tight',
            dpi=300)
```



Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code

# plot_documents_in_time_query_based_dataset(output_dir)
bib = BibliometricDataset("data/Resistance_QueryBased_Scopus.csv",
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fig.savefig(output_dir / "authors_ranking_reviews.png", bbox_inches='tight',
            dpi=300)

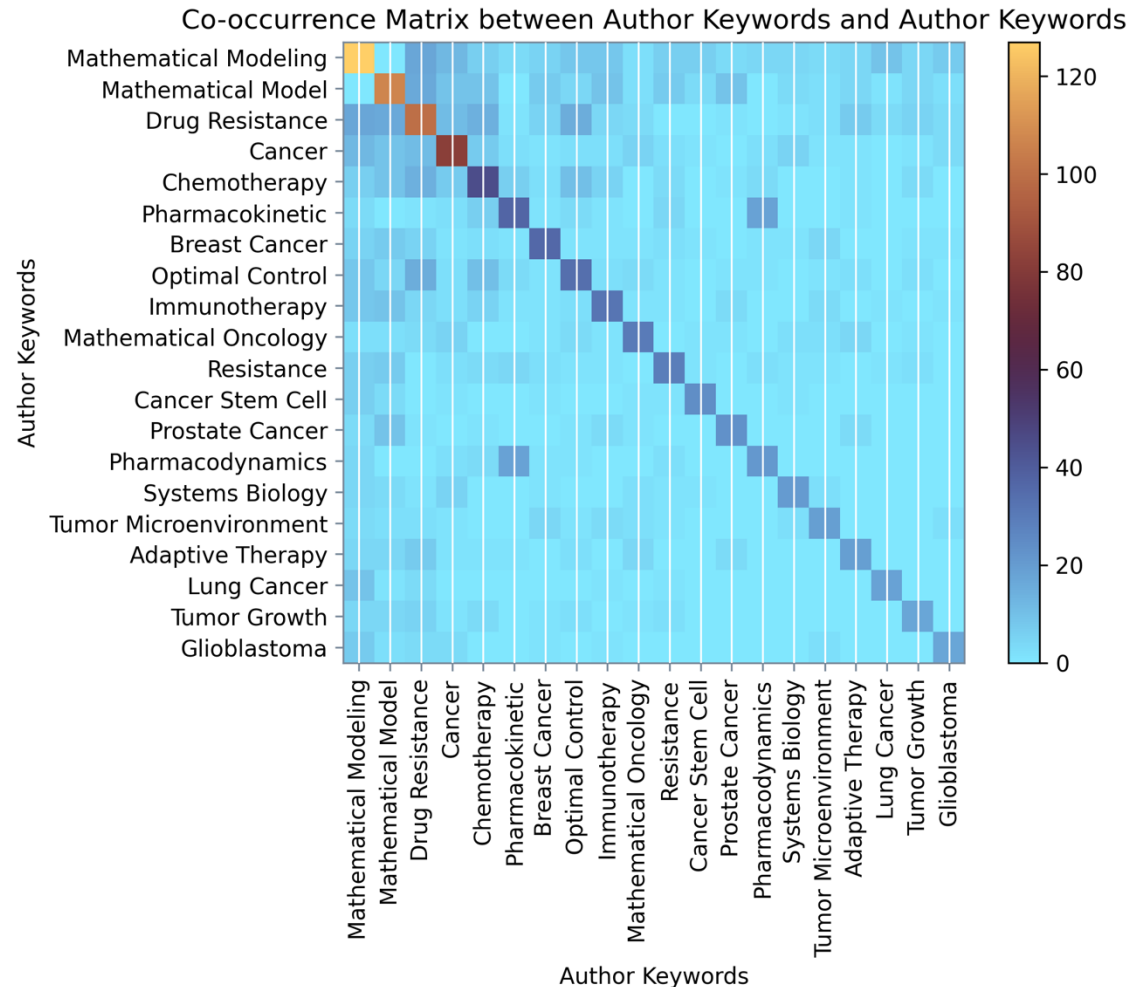
# see correlation between authors
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                             column_2="Author full names",
                                             top_n1=20,
                                             top_n2=20,
                                             ax=axes)

fig.savefig(output_dir / "authors_cooccurrence.png", bbox_inches='tight',
            dpi=300)

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fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.clean_author_keywords(threshold=0.9)
bib.plot_author_keywords_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "author_keywords_ranking.png", bbox_inches='tight',
            dpi=300)

# see co-occurrence of author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author Keywords",
                                             column_2="Author Keywords",
                                             top_n1=20,
                                             top_n2=20,
                                             ax=axes)

fig.savefig(output_dir / "author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)
```



Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code

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fig.savefig(output_dir / "authors_ranking.png", bbox_inches='tight',
            dpi=300)
fig.tight_layout()

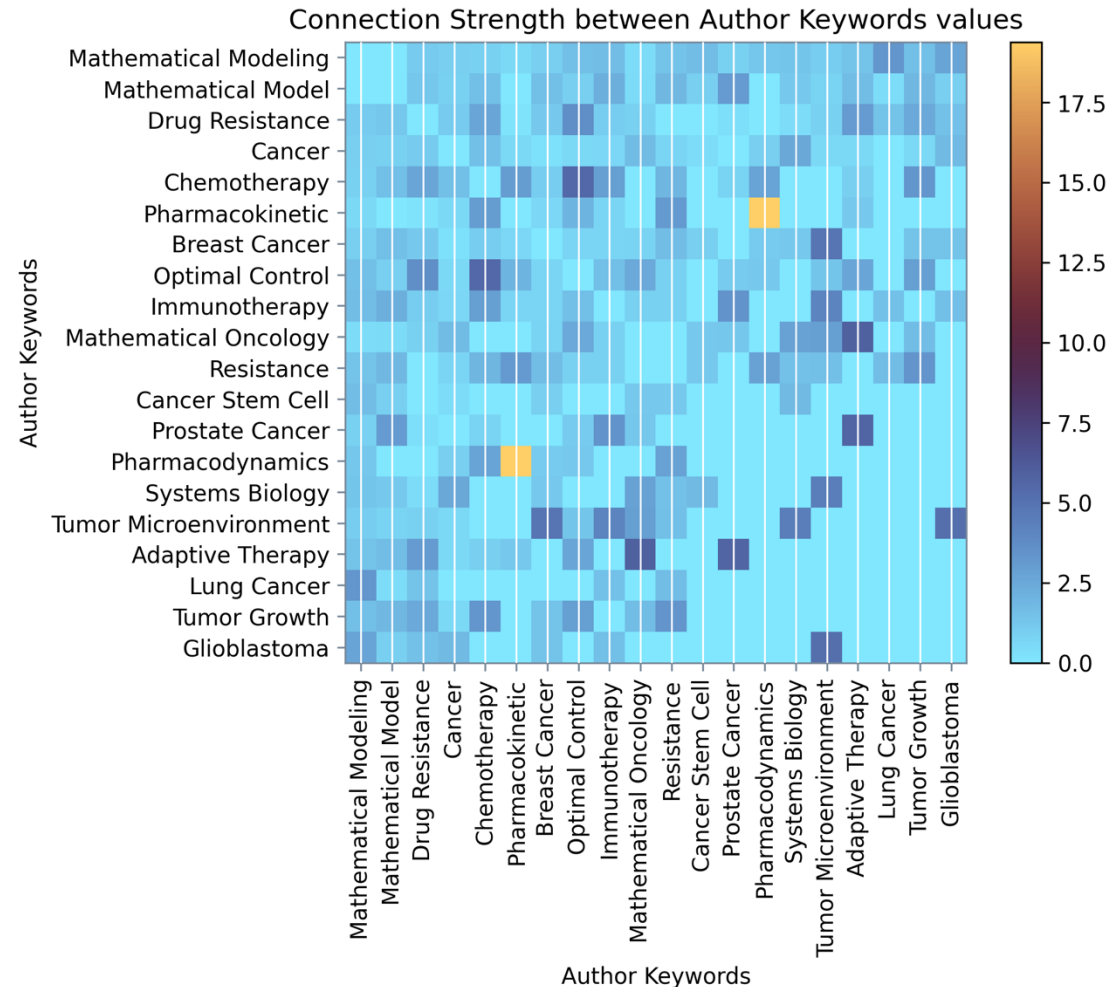
# do the same selecting reviews
bib_reviews = bib[bib["Document Type"] == "Review"]

fig, axes = plt.subplots(1, 1, figsize=(7, 5))
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bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                             column_2="Author full names",
                                             top_n1=20,
                                             top_n2=20,
                                             ax=axes)
fig.savefig(output_dir / "authors_cooccurrence.png", bbox_inches='tight',
            dpi=300)

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fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.clean_author_keywords(threshold=0.9)
bib.plot_author_keywords_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "author_keywords_ranking.png", bbox_inches='tight',
            dpi=300)

# see co-occurrence of author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author Keywords",
                                             column_2="Author Keywords",
                                             top_n1=20,
                                             top_n2=20,
                                             ax=axes)
fig.savefig(output_dir / "author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)
```



$$S = \frac{m \cdot c_{ij}}{c_{ii} \cdot c_{jj}}$$

Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```

fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib_reviews.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "authors_ranking_reviews.png", bbox_inches='tight',
            dpi=300)

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bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                           column_2="Author full names",
                                           top_n1=20,
                                           top_n2=20,
                                           ax=axes)
fig.savefig(output_dir / "authors_cooccurrence.png", bbox_inches='tight',
            dpi=300)

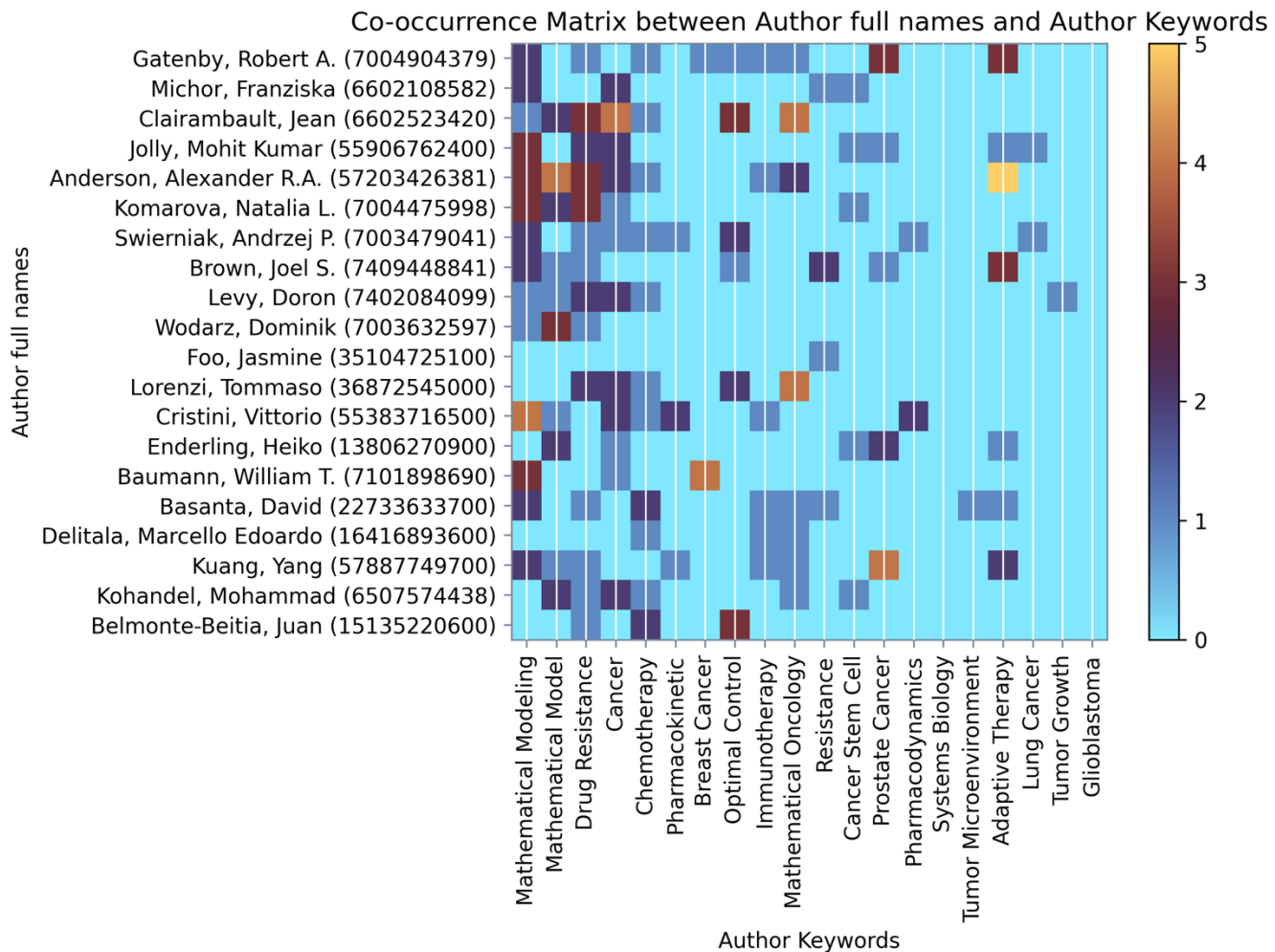
# see author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.clean_author_keywords(threshold=0.9)
bib.plot_author_keywords_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "author_keywords_ranking.png", bbox_inches='tight',
            dpi=300)

# see co-occurrence of author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author Keywords",
                                           column_2="Author Keywords",
                                           top_n1=20,
                                           top_n2=20,
                                           ax=axes)
fig.savefig(output_dir / "author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# see connection strength matrix of author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_connection_strength(column_1="Author Keywords",
                             column_2="Author Keywords",
                             top_n=20,
                             ax=axes)
fig.savefig(output_dir / "author_keywords_connection_strength.png", bbox_inches='tight',
            dpi=300)

# see co-occurrence between authors and author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                           column_2="Author Keywords",
                                           top_n1=20,
                                           top_n2=20,
                                           ax=axes)
fig.savefig(output_dir / "authors_author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

```



Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code
column_2="Author full names",
top_n1=20,
top_n2=20,
ax=axes)

fig.savefig(output_dir / "authors_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# see author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.clean_author_keywords(threshold=0.9)
bib.plot_author_keywords_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "author_keywords_ranking.png", bbox_inches='tight',
            dpi=300)

# see co-occurrence of author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author Keywords",
                                           column_2="Author Keywords",
                                           top_n1=20,
                                           top_n2=20,
                                           ax=axes)

fig.savefig(output_dir / "author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# see connection strength matrix of author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_connection_strength(column="Author Keywords",
                             top_n=20,
                             ax=axes)

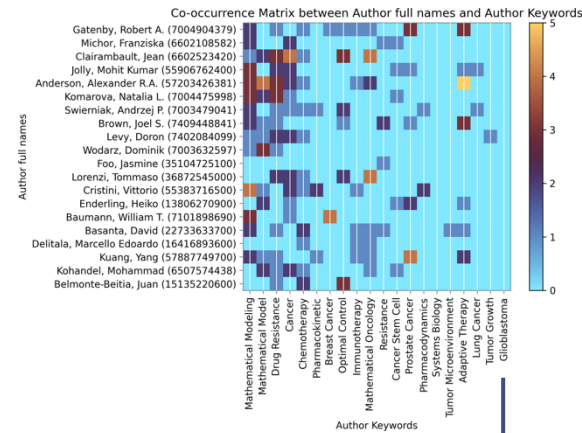
fig.savefig(output_dir / "author_keywords_connection_strength.png", bbox_inches='tight',
            dpi=300)

# see co-occurrence between authors and author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                           column_2="Author Keywords",
                                           top_n1=20,
                                           top_n2=20,
                                           ax=axes)

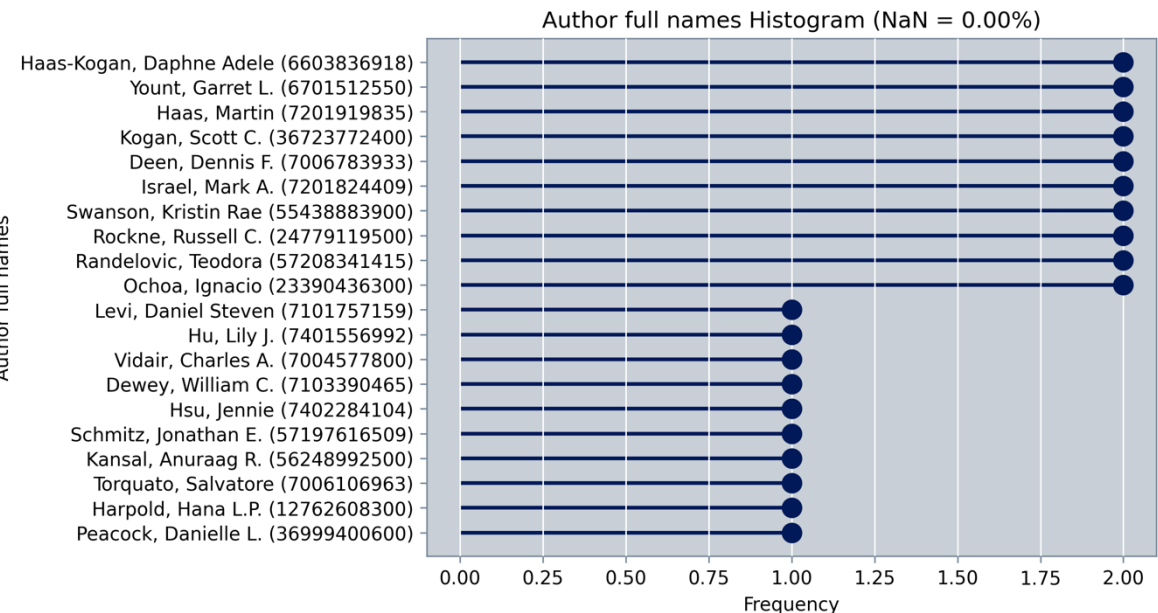
fig.savefig(output_dir / "authors_author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# select subset for glioblastoma
bib.glioblastoma = bib.bib["Author Keywords"].str.contains("glioblastoma", case=False, na=False)
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.glioblastoma.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "glioblastoma_authors_ranking.png", bbox_inches='tight',
            dpi=300)
fig.tight_layout()

# load regex dict
```



Author full names



Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code
column_2="Author full names",
top_n1=20,
top_n2=20,
ax=axes)

fig.savefig(output_dir / "authors_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# see author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.clean_author_keywords(threshold=0.9)
bib.plot_author_keywords_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "author_keywords_ranking.png", bbox_inches='tight',
            dpi=300)

# see co-occurrence of author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author Keywords",
                                           column_2="Author Keywords",
                                           top_n1=20,
                                           top_n2=20,
                                           ax=axes)
fig.savefig(output_dir / "author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# see connection strength matrix of author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_connection_strength(column="Author Keywords",
                             top_n=20,
                             ax=axes)
fig.savefig(output_dir / "author_keywords_connection_strength.png", bbox_inches='tight',
            dpi=300)

# see co-occurrence between authors and author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                           column_2="Author Keywords",
                                           top_n1=20,
                                           top_n2=20,
                                           ax=axes)
fig.savefig(output_dir / "authors_author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# select subset for glioblastoma
bib.glioblastoma = bib[bib["Author Keywords"].str.contains("glioblastoma", case=False, na=False)]
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.glioblastoma.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "glioblastoma_authors_ranking.png", bbox_inches='tight',
            dpi=300)
fig.tight_layout()

# load regex dict
```

Author Keywords are cool but:

- Often incomplete
- Limited choice
- Not all journals have them or share them

Solution: pattern matching with regular expression (regex)

- Flexible matching: e.g. “X.ray” matches X ray and X-ray

Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code
fig.savefig(output_dir / "author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

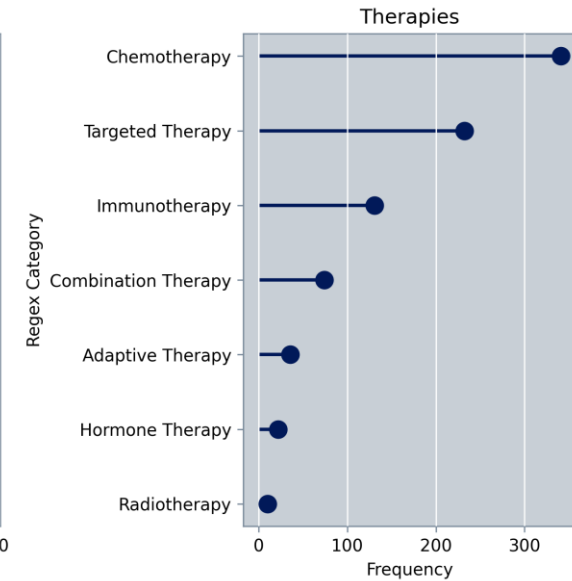
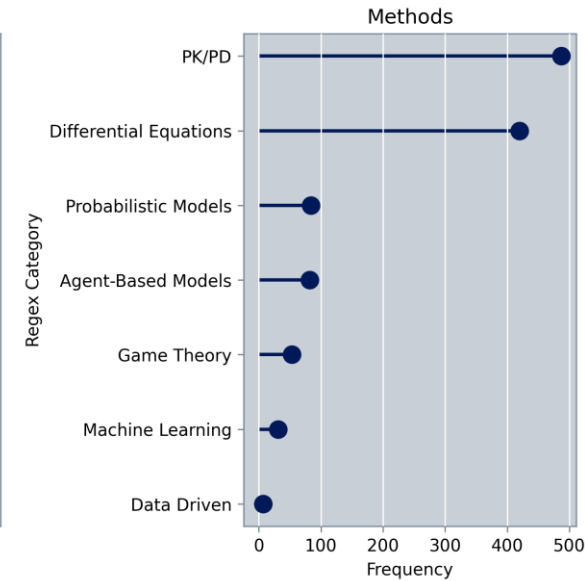
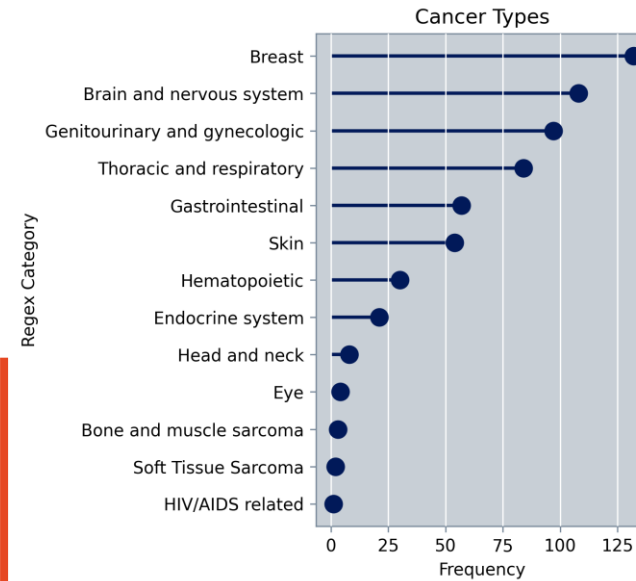
# see connection strength matrix of author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
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                             top_n=20,
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            dpi=300)

# see co-occurrence between authors and author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                             column_2="Author Keywords",
                                             top_n1=20,
                                             top_n2=20,
                                             ax=axes)
fig.savefig(output_dir / "authors_author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# select subset for glioblastoma
bib.glioblastoma = bib[bib["Author Keywords"].str.contains("glioblastoma", case=False, na=False)]
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.glioblastoma.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "glioblastoma_authors_ranking.png", bbox_inches='tight',
            dpi=300)
fig.tight_layout()

# load regex dict
with open("config/regex/cancer_types.json", "r") as f:
    cancer_types = json.load(f)
with open("config/regex/methods.json", "r") as f:
    methods = json.load(f)
with open("config/regex/therapies.json", "r") as f:
    therapies = json.load(f)

# plot cancer types ranking
fig, axes = plt.subplots(1, 3, figsize=(15, 5))
bib.plot_regex_categories_ranking(cancer_types, top_n=20, ax=axes[0])
bib.plot_regex_categories_ranking(methods, top_n=20, ax=axes[1])
bib.plot_regex_categories_ranking(therapies, top_n=20, ax=axes[2])
axes[0].set_title("Cancer Types")
axes[1].set_title("Methods")
axes[2].set_title("Therapies")
fig.tight_layout()
fig.savefig(output_dir / "regex_categories_ranking.png", bbox_inches='tight',
            dpi=300)
```



Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code

fig.savefig(output_dir / "author_keywords_connection_strength.png", bbox_inches='tight',
            dpi=300)

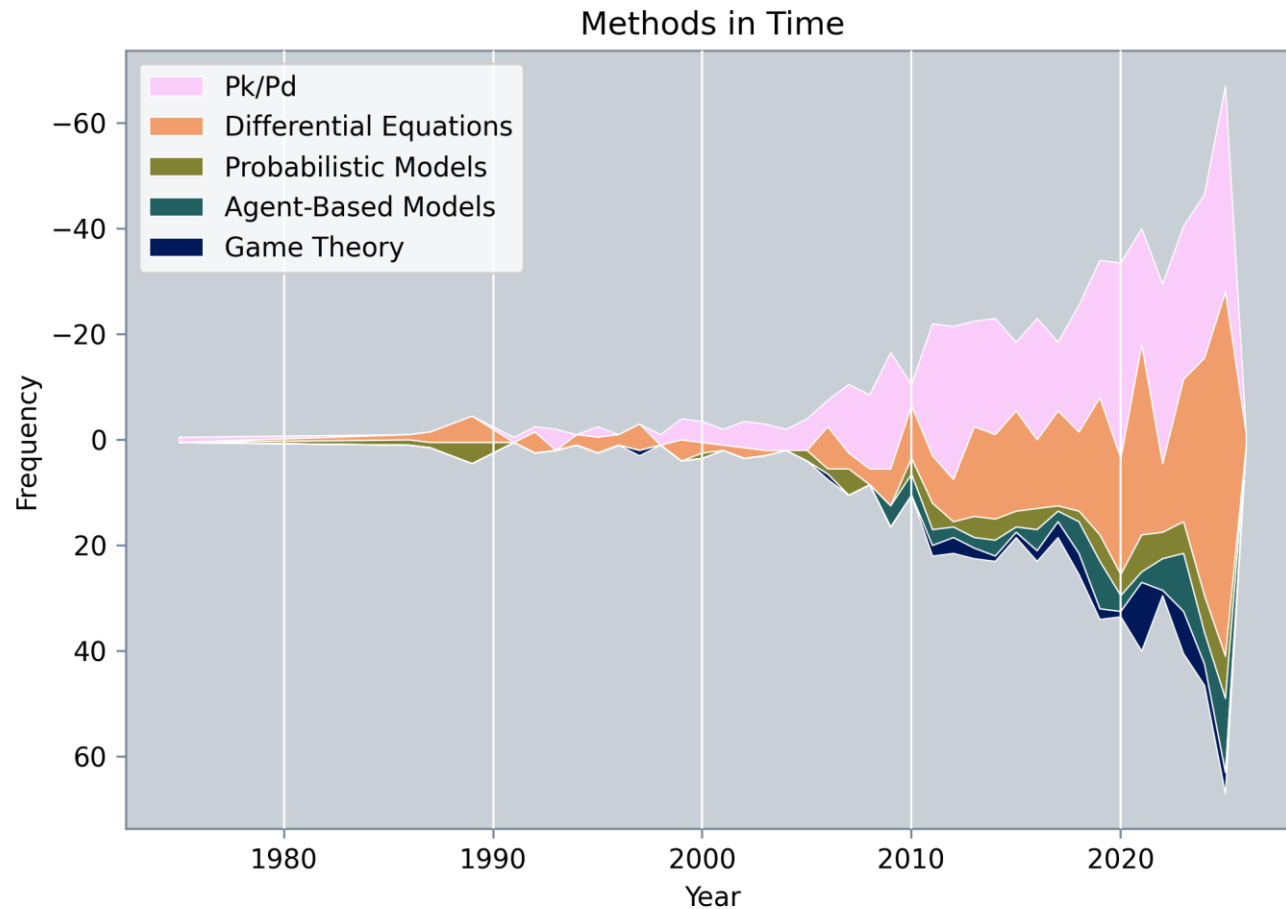
# see co-occurrence between authors and author keywords
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_column_column_cooccurrence_matrix(column_1="Author full names",
                                           column_2="Author Keywords",
                                           top_n1=20,
                                           top_n2=20,
                                           ax=axes)
fig.savefig(output_dir / "authors_author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# select subset for glioblastoma
bib_glioblastoma = bib[bib["Author Keywords"].str.contains("glioblastoma", case=False, na=False)]
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib_glioblastoma.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "glioblastoma_authors_ranking.png", bbox_inches='tight',
            dpi=300)
fig.tight_layout()

# load regex dict
with open("config/regex/cancer_types.json", "r") as f:
    cancer_types = json.load(f)
with open("config/regex/methods.json", "r") as f:
    methods = json.load(f)
with open("config/regex/therapies.json", "r") as f:
    therapies = json.load(f)

# plot cancer types ranking
fig, axes = plt.subplots(1, 3, figsize=(15, 5))
bib.plot_regex_categories_ranking(cancer_types, top_n=20, ax=axes[0])
bib.plot_regex_categories_ranking(methods, top_n=20, ax=axes[1])
bib.plot_regex_categories_ranking(therapies, top_n=20, ax=axes[2])
axes[0].set_title("Cancer Types")
axes[1].set_title("Methods")
axes[2].set_title("Therapies")
fig.tight_layout()
fig.savefig(output_dir / "regex_categories_ranking.png", bbox_inches='tight',
            dpi=300)

# plot regex categories in time
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_regex_categories_in_time(methods, top_n=5, ax=axes)
axes.set_title("Methods in Time")
axes.legend(loc='upper left', fontsize=10)
fig.tight_layout()
fig.savefig(output_dir / "methods_in_time.png", bbox_inches='tight',
            dpi=300)
```



Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code
fig.savefig(output_dir / "authors_author_keywords_cooccurrence.png", bbox_inches='tight',
            dpi=300)

# select subset for glioblastoma
bib_glioblastoma = bib[bib["Author Keywords"].str.contains("glioblastoma", case=False, na=False)]
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib_glioblastoma.plot_authors_ranking(top_n=20, ax=axes)
fig.savefig(output_dir / "glioblastoma_authors_ranking.png", bbox_inches='tight',
            dpi=300)
fig.tight_layout()

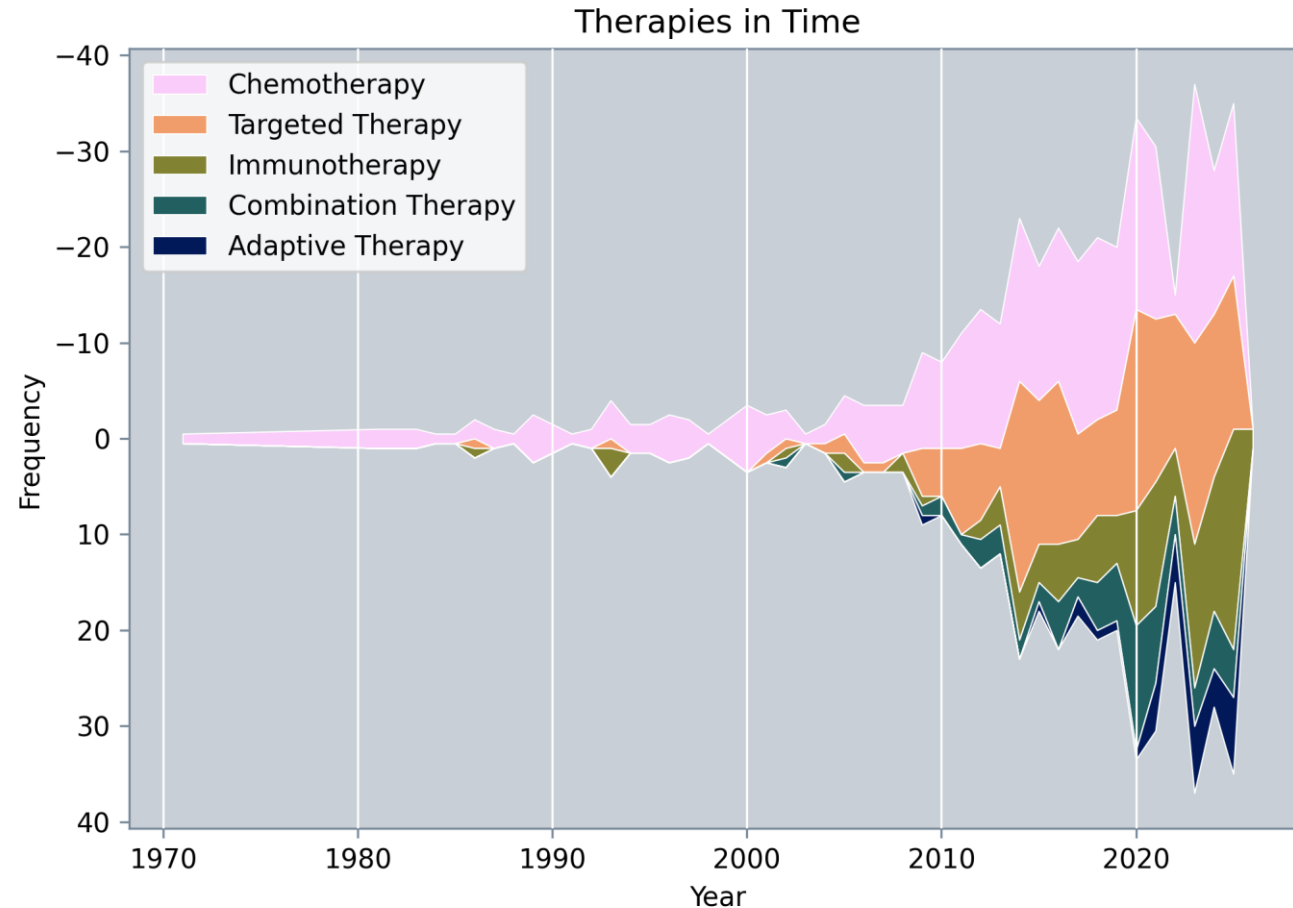
# load regex dict
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    cancer_types = json.load(f)
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    methods = json.load(f)
with open("config/regex/therapies.json", "r") as f:
    therapies = json.load(f)

# plot cancer types ranking
fig, axes = plt.subplots(1, 3, figsize=(15, 5))
bib.plot_regex_categories_ranking(cancer_types, top_n=20, ax=axes[0])
bib.plot_regex_categories_ranking(methods, top_n=20, ax=axes[1])
bib.plot_regex_categories_ranking(therapies, top_n=20, ax=axes[2])
axes[0].set_title("Cancer Types")
axes[1].set_title("Methods")
axes[2].set_title("Therapies")
fig.tight_layout()
fig.savefig(output_dir / "regex_categories_ranking.png", bbox_inches='tight',
            dpi=300)

# plot regex categories in time
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_regex_categories_in_time(methods, top_n=5, ax=axes)
axes.set_title("Methods in Time")
axes.legend(loc='upper left', fontsize=10)
fig.tight_layout()
fig.savefig(output_dir / "methods_in_time.png", bbox_inches='tight',
            dpi=300)

fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_regex_categories_in_time(therapies, top_n=5, ax=axes)
axes.set_title("Therapies in Time")
axes.legend(loc='upper left', fontsize=10)
fig.tight_layout()
fig.savefig(output_dir / "therapies_in_time.png", bbox_inches='tight',
            dpi=300)

# plot custom regex
```



Bibliometrics: see authors

Example: Mathematical Modelling in Cancer Resistance

```
Code
fig.tight_layout()
fig.savefig(output_dir / "regex_categories_ranking.png", bbox_inches='tight',
            dpi=300)

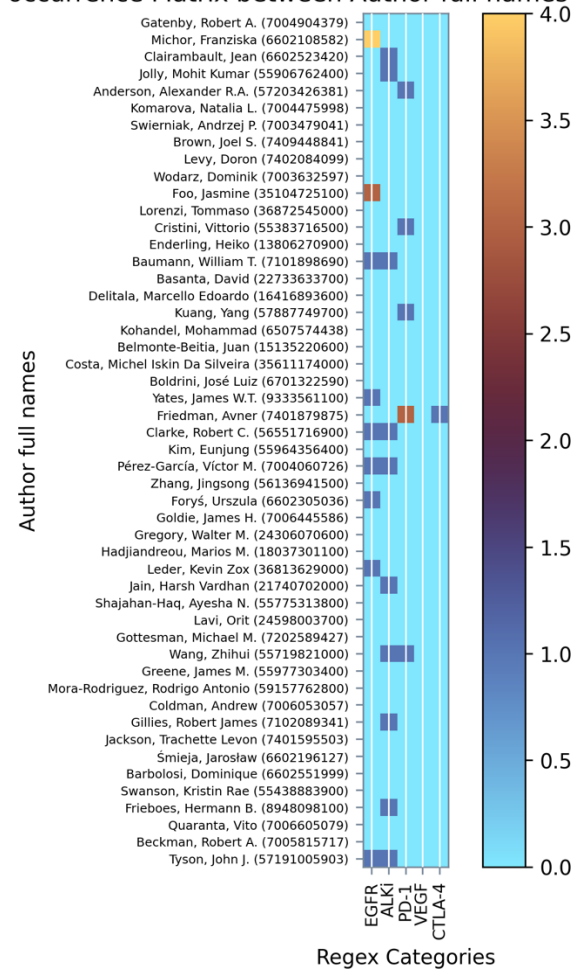
# plot regex categories in time
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_regex_categories_in_time(methods, top_n=5, ax=axes)
axes.set_title("Methods in Time")
axes.legend(loc='upper left', fontsize=10)
fig.tight_layout()
fig.savefig(output_dir / "methods_in_time.png", bbox_inches='tight',
            dpi=300)

fig, axes = plt.subplots(1, 1, figsize=(7, 5))
bib.plot_regex_categories_in_time(therapies, top_n=5, ax=axes)
axes.set_title("Therapies in Time")
axes.legend(loc='upper left', fontsize=10)
fig.tight_layout()
fig.savefig(output_dir / "therapies_in_time.png", bbox_inches='tight',
            dpi=300)

# plot custom regex
fig, axes = plt.subplots(1, 1, figsize=(7, 5))
custom_regex = {
    "EGFR": [r"EGFR|Epidermal Growth Factor Receptor"],
    "VEGF": [r"VEGF|Vascular Endothelial Growth Factor"],
    "PD-1": [r"PD-1|Programmed cell death protein 1", r"PD-L1|Programmed death-ligand 1"],
    "CTLA-4": [r"CTLA-4|Cytotoxic T-lymphocyte-associated protein 4"],
    "ALKI": [r"ALK|Anaplastic Lymphoma Kinase Inhibitor"],
}
bib.plot_regex_categories_ranking(custom_regex, top_n=20, ax=axes)
fig.tight_layout()
fig.savefig(output_dir / "custom_regex_ranking.png", bbox_inches='tight',
            dpi=300)

# plot custom regex cooccurrence
fig, axes = plt.subplots(1, 1, figsize=(5, 7))
bib.plot_column_regex_cooccurrence_matrix(column="Author full names",
                                          regex_dict=custom_regex,
                                          top_n1=50,
                                          top_n2=20,
                                          ax=axes,
                                          categories=True)
axes.set_xticklabels(axes.get_xticklabels(), rotation=90, fontsize=8)
axes.set_yticklabels(axes.get_yticklabels(), fontsize=6)
fig.tight_layout()
fig.savefig(output_dir / "custom_regex_cooccurrence.png", bbox_inches='tight',
            dpi=300)
```

Co-occurrence Matrix between Author full names and Regex Categories



- Introduction
- Staying updated with web scraping
- Exploring the literature with Bibliometrics
- Integrations with Large Language Models

Large Language Models are having an immense impact in our daily life and in science

Article

The Virtual Lab of AI agents designs new SARS-CoV-2 nanobodies

<https://doi.org/10.1038/s41586-025-09442-9> Kyle Swanson¹, Wesley Wu², Nash L. Bulaong², John E. Pak^{2,3} & James Zou^{1,2,3}

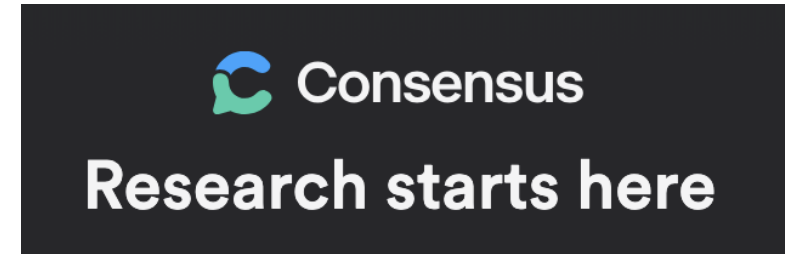
Received: 26 November 2024

Article | [Open access](#) | Published: 04 February 2026

Synthesizing scientific literature with retrieval-augmented language models

[Akari Asai](#), [Jacqueline He](#), [Rulin Shao](#), [Weijia Shi](#), [Amanpreet Singh](#), [Joseph Chee Chang](#), [Kyle Lo](#), [Luca Soldaini](#), [Sergey Feldman](#), [Mike D'Arcy](#), [David Wadden](#), [Matt Latzke](#), [Jenna Sparks](#), [Jena D. Hwang](#), [Varsha Kishore](#), [Minyang Tian](#), [Pan Ji](#), [Shengyan Liu](#), [Hao Tong](#), [Bohao Wu](#), [Yanyu Xiong](#), [Luke Zettlemoyer](#), [Graham Neubig](#), [Daniel S. Weld](#), ... [Hannaneh Hajishirzi](#) ✉ [+ Show authors](#)

[Nature](#) (2026) | [Cite this article](#)



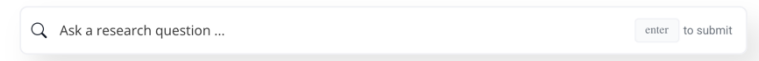
[Link](#)



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AI Search Engine For Scientific Research

Cut twenty hours of research down to just twenty minutes



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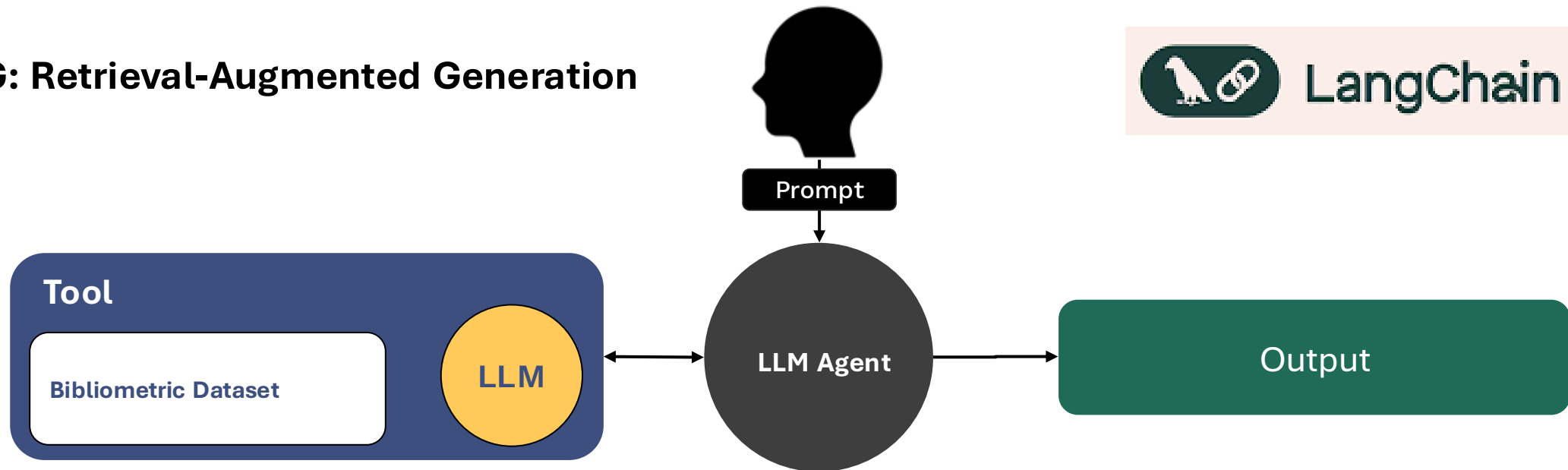
Large Language Models are having an immense impact in our daily life and in science

We're experimenting our owns:

Large Language Models are having an immense impact in our daily life and in science

We're experimenting our owns:

RAG: Retrieval-Augmented Generation



"LiRA: Literature Review Automated"

🔗 Automate your weekly scientific literature review

LiRA is a CLI Python program based on PyMed and SerpAPI to search on PubMed and get the results programmatically in a readable HTML page.

I created LiRA mainly for myself, but feel free to use it if you find it useful.



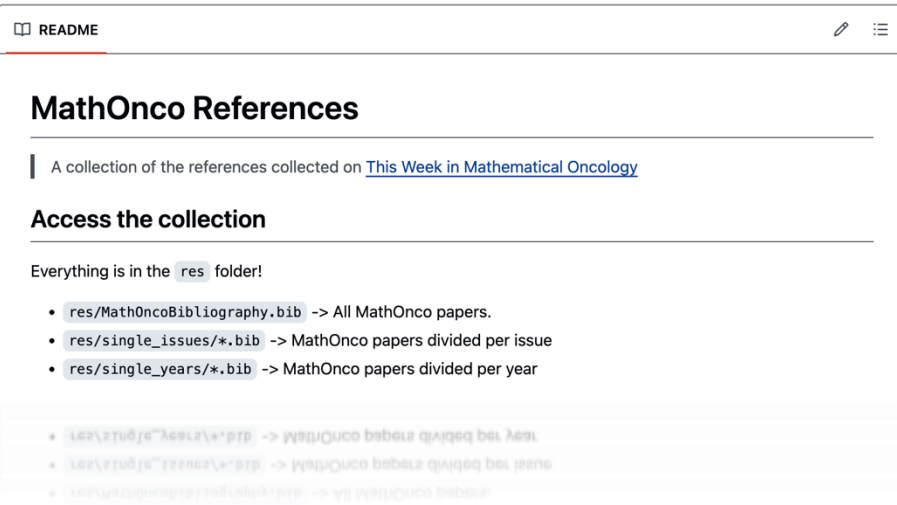
Can an LLM discriminate **"interesting"** publications?

"LiRA: Literature Review Automated"

Automate your weekly scientific literature review

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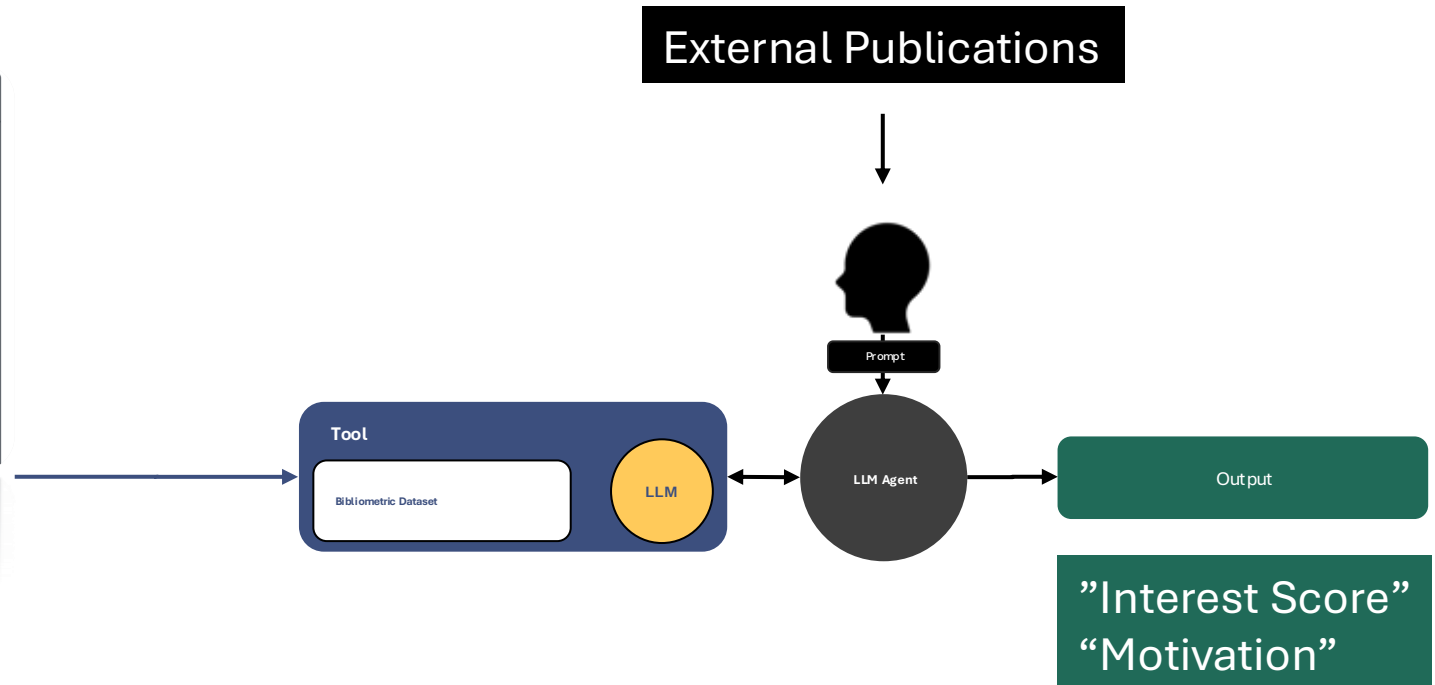
I created LiRA mainly for myself, but feel free to use it if you find it useful.



TWiMO Dataset

- <https://github.com/MathOnco/Newsletter-Bibliography>

Can an LLM discriminate "interesting" publications?

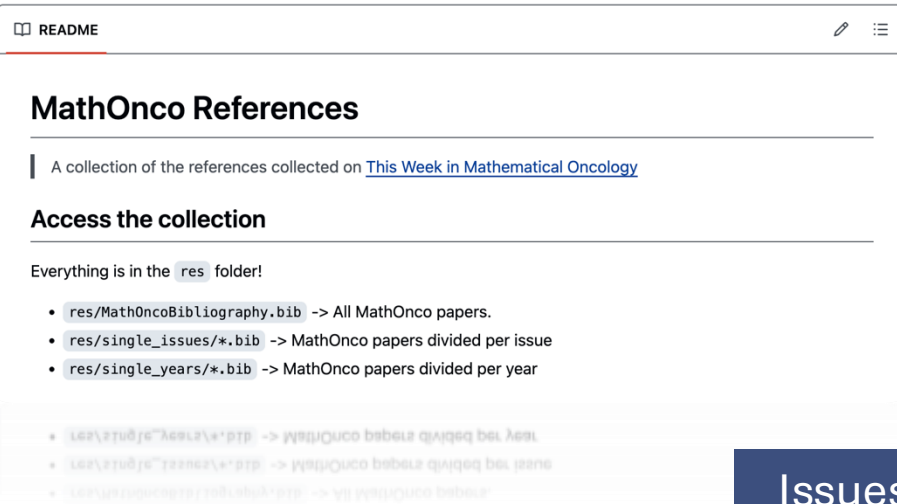


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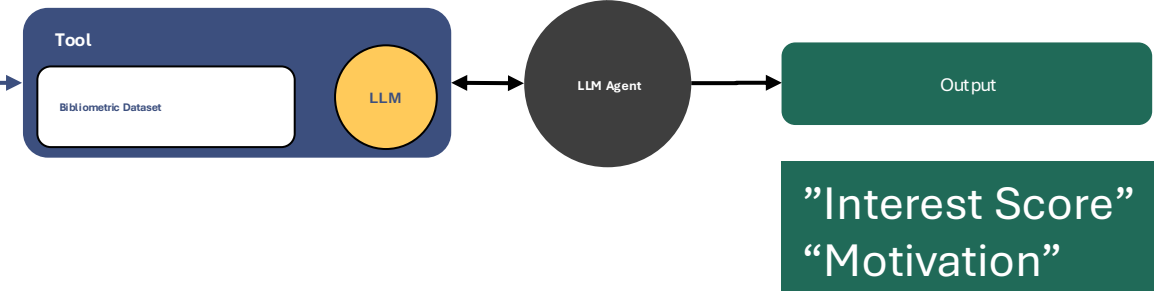
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Issues 1 - 289

Can an LLM discriminate "interesting" publications?

LiRA publications in the time frame of issues 290-299



TWiMO Dataset

- <https://github.com/MathOnco/Newsletter-Bibliography>

"LiRA: Literature Review Automated"

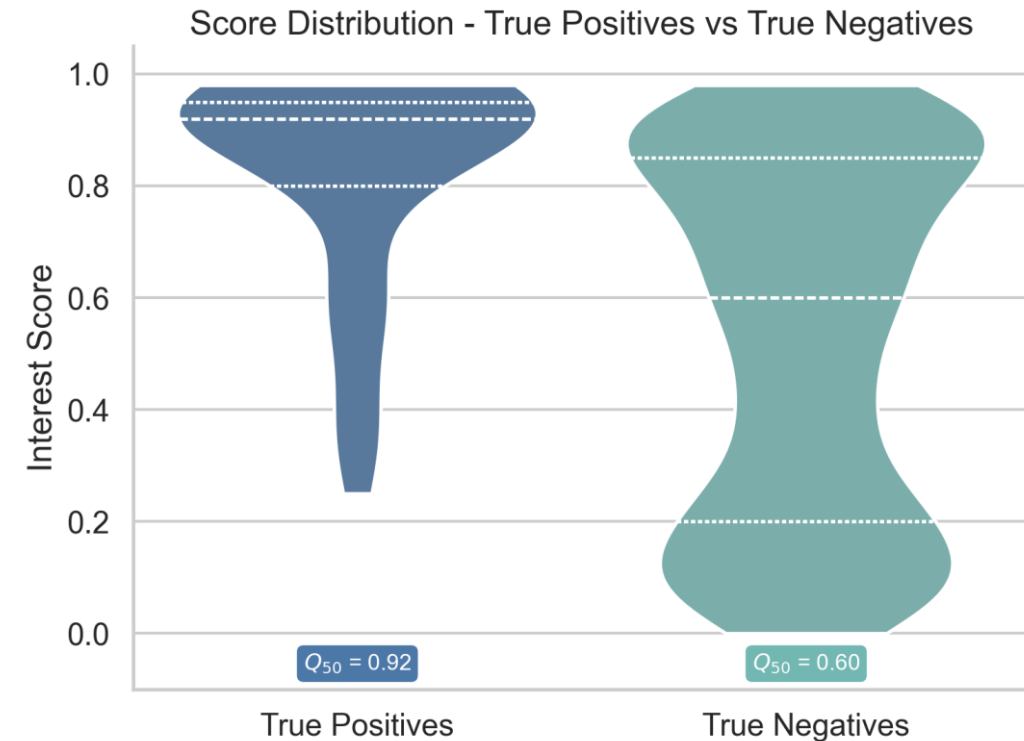
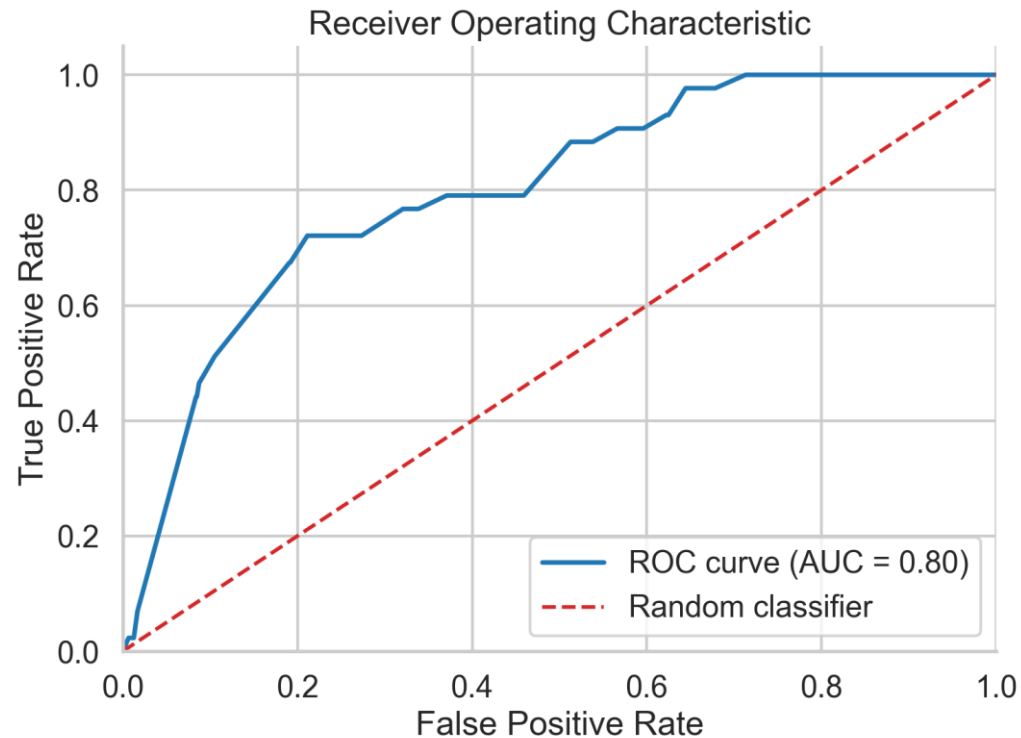
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Can an LLM discriminate "interesting" publications?

> [Breast Cancer Res Treat](#). 2024 Jul;206(1):91-103. doi: 10.1007/s10549-024-07273-2. Epub 2024 May 5.

Treatment patterns and clinical outcomes in patients with metastatic triple-negative breast cancer: a large-scale data analysis using the Japanese claims database



[Takayuki Kimura](#)¹, [Tomoko Takami](#)², [Yi Piao](#)², [Ioanna Ntalla](#)³, [Shigehira Saji](#)⁴

Affiliations + expand

PMID: 38704772 PMCID: [PMC11182808](#) DOI: [10.1007/s10549-024-07273-2](#)



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View Complete Issue

Score: **0.25**

Motivation: "Large-scale real-world data analysis in oncology is useful background but the paper is primarily clinical/epidemiological rather than methodological or modelling-focused."

> [PLoS Comput Biol](#). 2024 May 3;20(5):e1012088. doi: 10.1371/journal.pcbi.1012088. eCollection 2024 May.

Machine-learning and mechanistic modeling of metastatic breast cancer after neoadjuvant treatment



[Sebastien Benzekry](#)¹, [Michalis Mastroianni](#)², [Chiara Nicolò](#)³, [John M L Ebos](#)^{2, 4}

Affiliations + expand

PMID: 38701089 PMCID: [PMC11095706](#) DOI: [10.1371/journal.pcbi.1012088](#)



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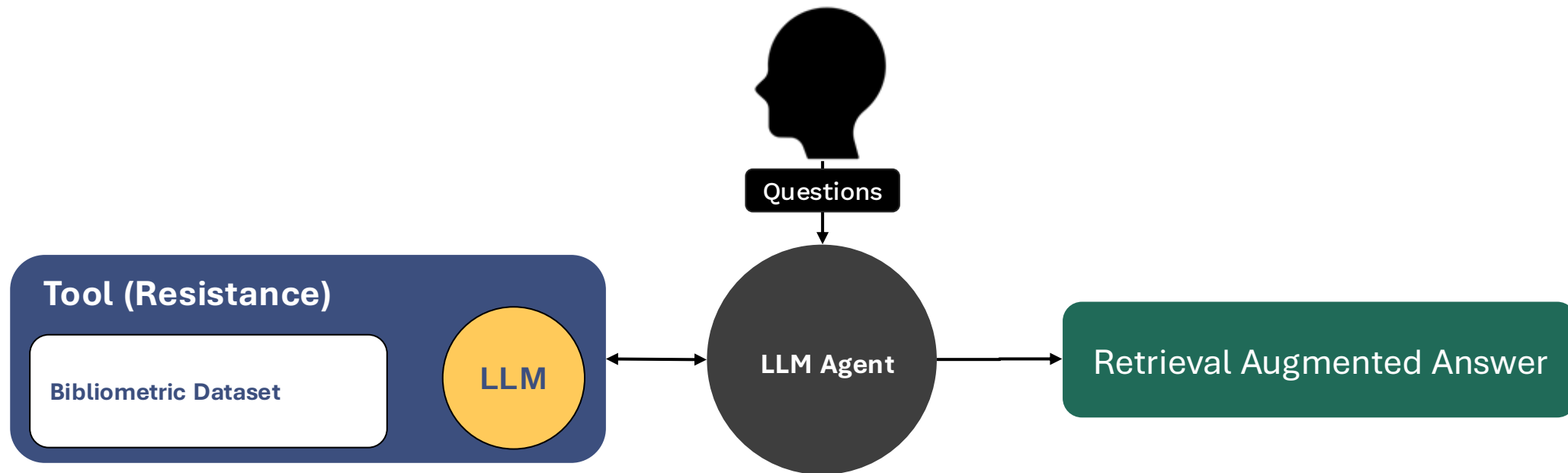


View Complete Issue

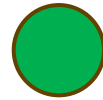
Score: **0.95**

Motivation: "Exemplary combination of mechanistic PK/PD-style modeling, non-linear mixed-effects, ML and public data for metastatic dynamics - directly in TWIMO's methodological wheelhouse."

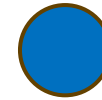
Direct use with bibliometrics data



Tested*:



Accuracy = Sentence Score / Number of Sentences



Hallucinations (N references)

Q1: “What is adaptive therapy?”

Ours

0.99

0/2

Hi Franco
Where should we start?

0.99

0/5

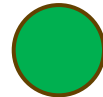
Consensus

Research starts here

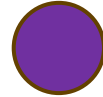
1

0/20

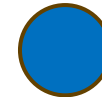
Tested*:



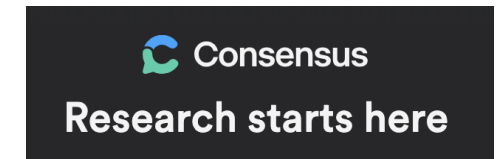
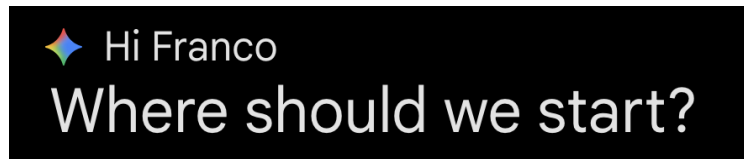
Accuracy = Sentence Score / Number of Sentences



In scope (True/False)



Hallucinations (N references)



Q2: “Provide me 5 relevant references about mathematical modeling of cancer resistance using experimental preclinical data.”

1

0/5

4/5

0.4

1/5

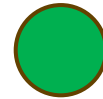
4/5

1

0/5

4/5

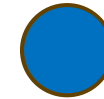
Tested*:



Accuracy = Sentence Score / Number of Sentences



In scope (True/False)



Hallucinations (N references)

Ours

Q3: “Provide me 5 relevant references about mathematical modeling of cancer resistance using mechanistic learning.”

0.4

1/5

1/5

Hi Franco
Where should we start?

0.6

2/5

2/5

Consensus

Research starts here

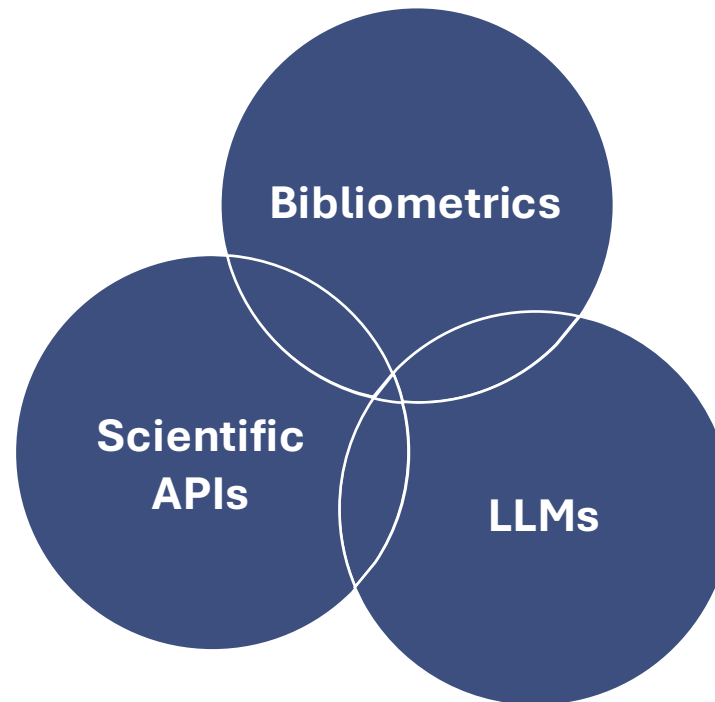
1

0/5

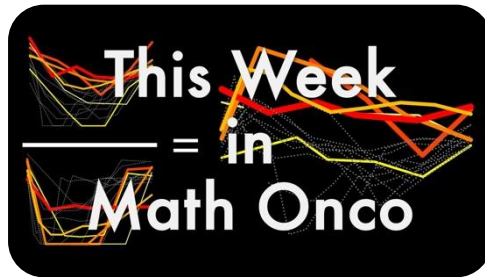
4/5

“Much of a scientist’s work consists of reading and writing.”

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Acknowledgments



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David Basanta
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Thank you!

Announcement: IMO Latte Art 2026

*By AACR: Atlantic Association for
Cappuccino Representation*